Lesson 1

AIR-CONDITIONING SYSTEMS

Electricity Tariffs

ELECTRICITY TARIFFS FROM 1 OCTOBER 2024

	Existing Tariff (without GST)	New Tariff (without GST)	New Tariff (with 9% GST)
LOW TENSION SUPPLIES, DOMESTIC All units, ¢/kWh	29.88	29.10	31.72
LOW TENSION SUPPLIES, NON-DOMESTIC All units, ¢/kWh	29.88	29.10	31.72
HIGH TENSION SMALL (HTS) SUPPLIES Contracted Capacity Charge $/kW/month	16.37	16.37	17.84
Uncontracted Capacity Charge $/chargeable kW/month	24.56	24.56	26.77
kWh charge, ¢/kWh			
Peak period (7.00am to 11.00pm)	27.05	26.46	28.84
Off-peak period (11.00pm to 7.00am)	17.38	16.16	17.61
Reactive power Charge ¢/chargeable kVARh	0.59	0.59	0.64

Electricity Tariffs

ELECTRICITY TARIFFS FROM 1 OCTOBER 2024

	Existing Tariff (without GST)	New Tariff (without GST)	New Tariff (with 9% GST)
HIGH TENSION LARGE (HTL) SUPPLIES			
Contracted Capacity Charge $/kW/month	16.37	16.37	17.84
Uncontracted Capacity Charge $/chargeable kW/month	24.56	24.56	26.77
kWh charge, ¢/kWh			
Peak period (7.00am to 11.00pm)	26.83	26.24	28.60
Off-peak period (11.00pm to 7.00am)	17.37	16.15	17.60
Reactive power Charge ¢/chargeable kVARh	0.59	0.59	0.64
EXTRA HIGH TENSION (EHT) SUPPLIES			
Contracted Capacity Charge $/kW/month	12.90	12.90	14.06
Uncontracted Capacity Charge $/chargeable kW/month	19.35	19.35	21.09
kWh charge, ¢/kWh			
Peak period (7.00am to 11.00pm)	25.88	25.28	27.56
Off-peak period (11.00pm to 7.00am)	17.27	16.05	17.49
Reactive power Charge ¢/chargeable kVARh	0.48	0.48	0.52

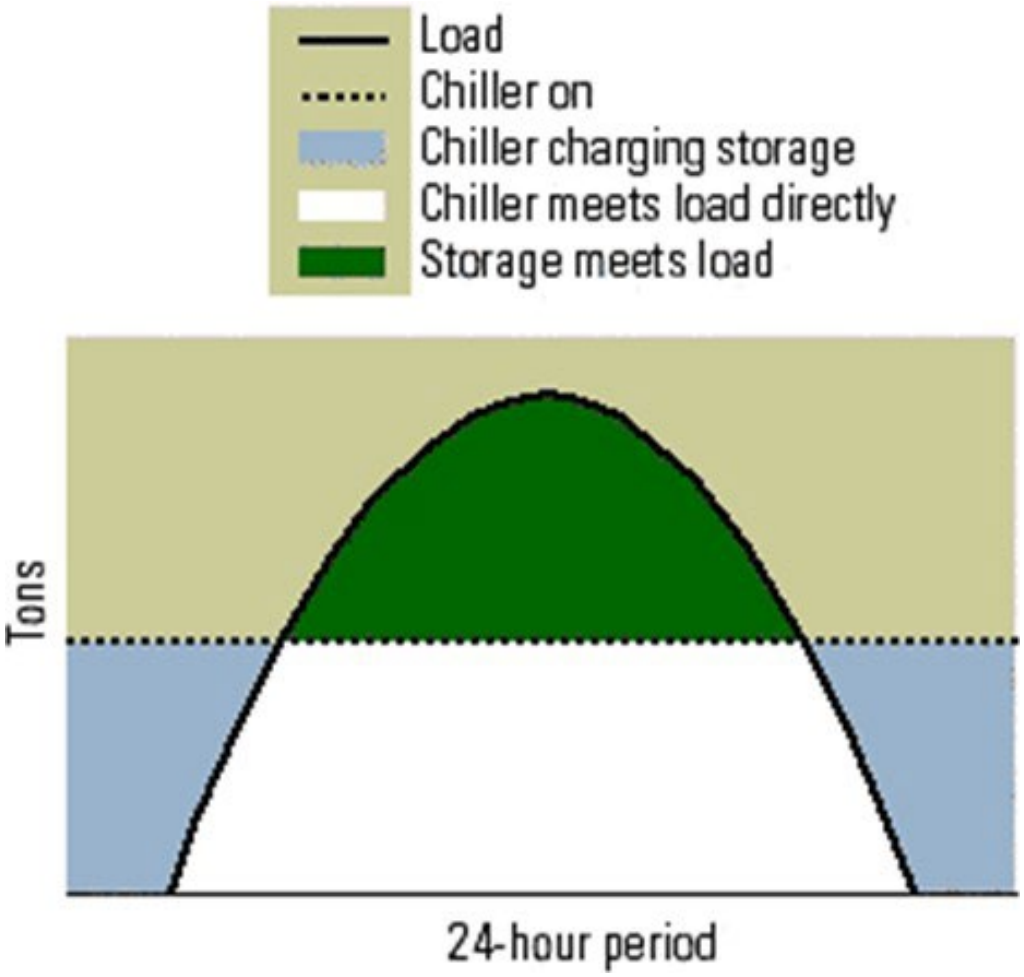
Thermal Energy Storage Systems

During Night

When tariffs are lower, chilled water or ice stored in tanks

During Day

When cooling is needed, chilled water drawn from storage tank

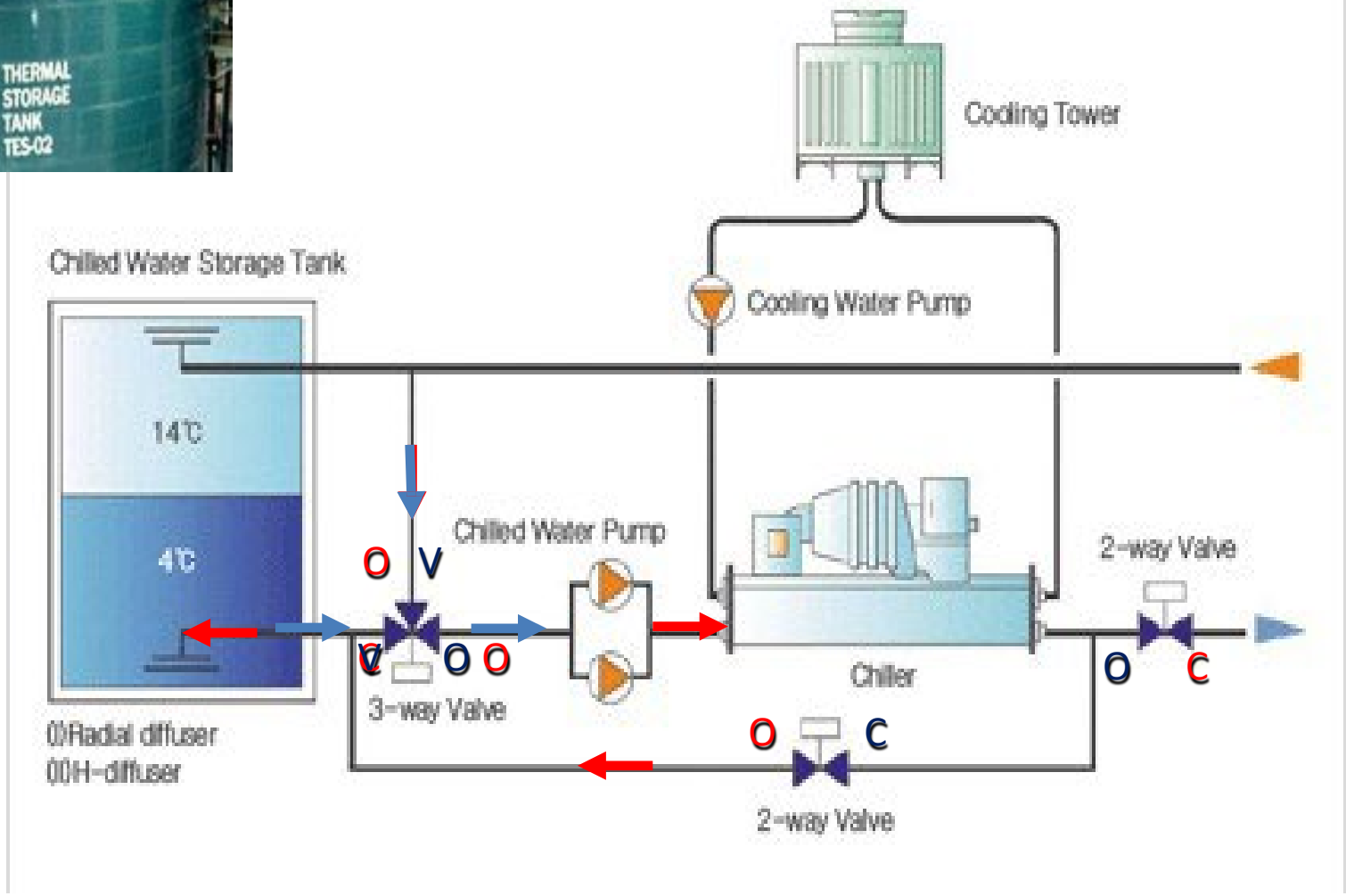


Thermal Energy Storage Systems

- Therefore, TES should be considered when:
- A facility's cooling needs vary substantially during the day.
- A facility's maximum cooling load is much greater than the average load
- The utility rate structure has high demand charges or a high differential between on- and off-peak energy rates

Thermal Energy Storage Systems

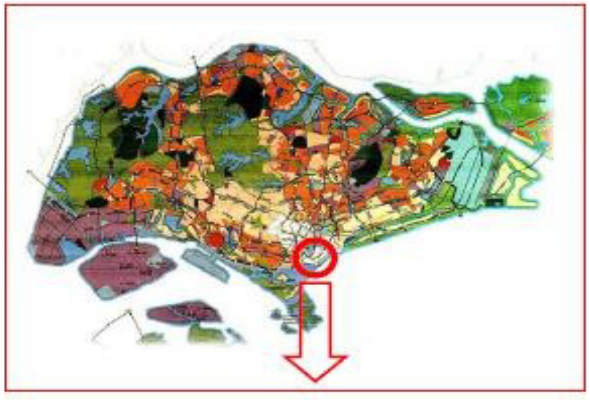


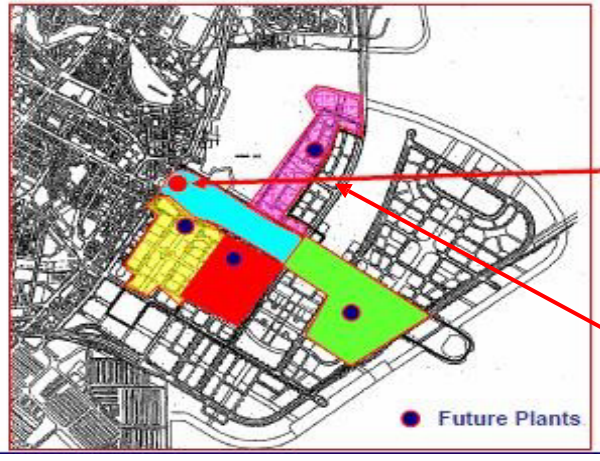


Singapore District Cooling

District Cooling in Marina Bay

SDC





Master Plan

- > 8 million m^2 GFA
- > 900 MW_r cooling load
- 5 district cooling plants



Location of 1st plant : One Raffles Quay

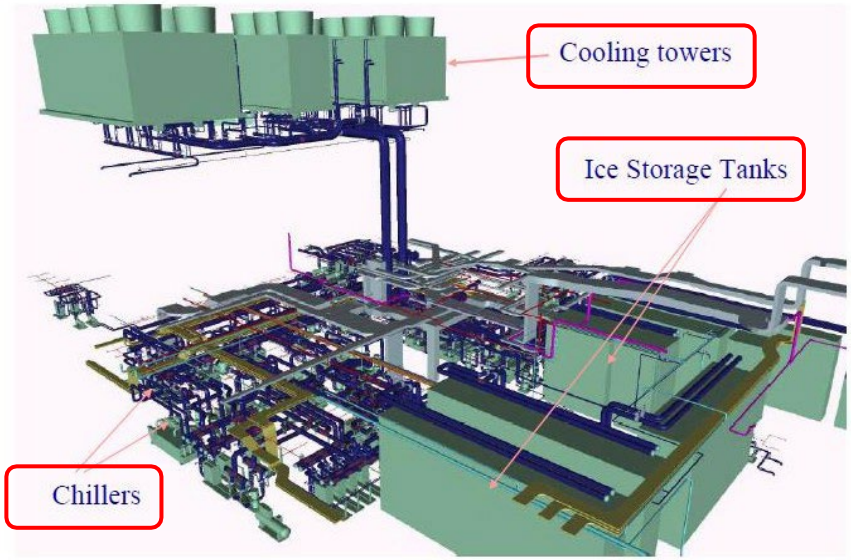
Planned: 150 MW_r

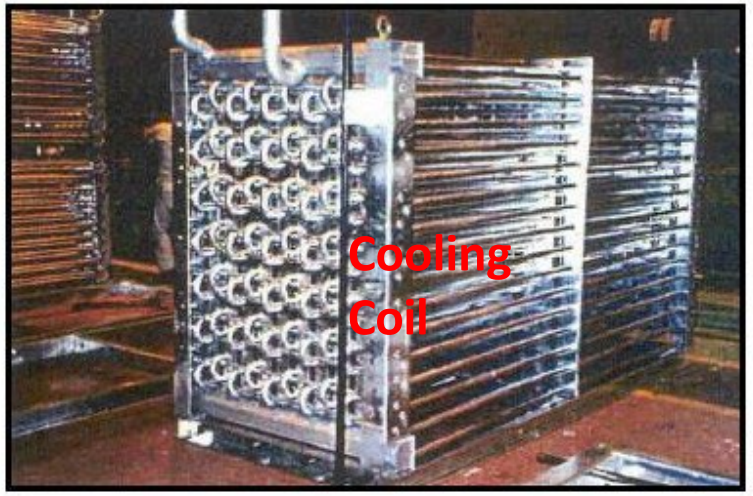
Installed: 57 MW_r

Cooling Towers

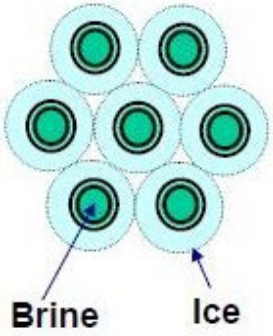


Singapore District Cooling









Planned Capacity	150 MWr
Current installed capacity	57 MWr

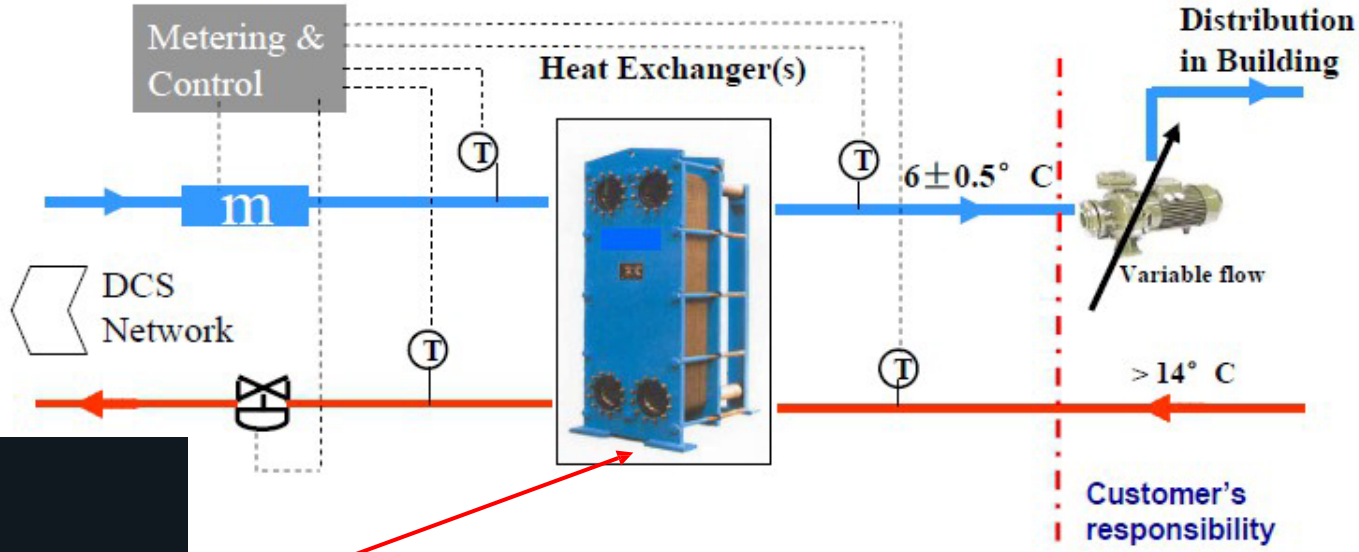
Key Items	Capacity	Ultimate Qty (Unit)	Currently Installed (Unit)
Water chillers	3 MWr	1	1
Water chillers	7 MWr	4	2
Brine chillers	10 MWr	6	2
Ice Storage subsystems		6	2
Discharge capacity	10 MWr		
Thermal storage	80 MWrh*		
Cooling towers	5 MWr	23	10

Singapore District Cooling

Supply Connection & Interface

SDC

- Service provided via heat exchangers
- "Variable flow" design for downstream reticulation
- Supply temperature regulated at 6°C±0·5°C
- Chilled water return temperature to be > 14°C



Keppel DHCS



Changi Business Park



Biopolis@One North



Mediapolis@One North



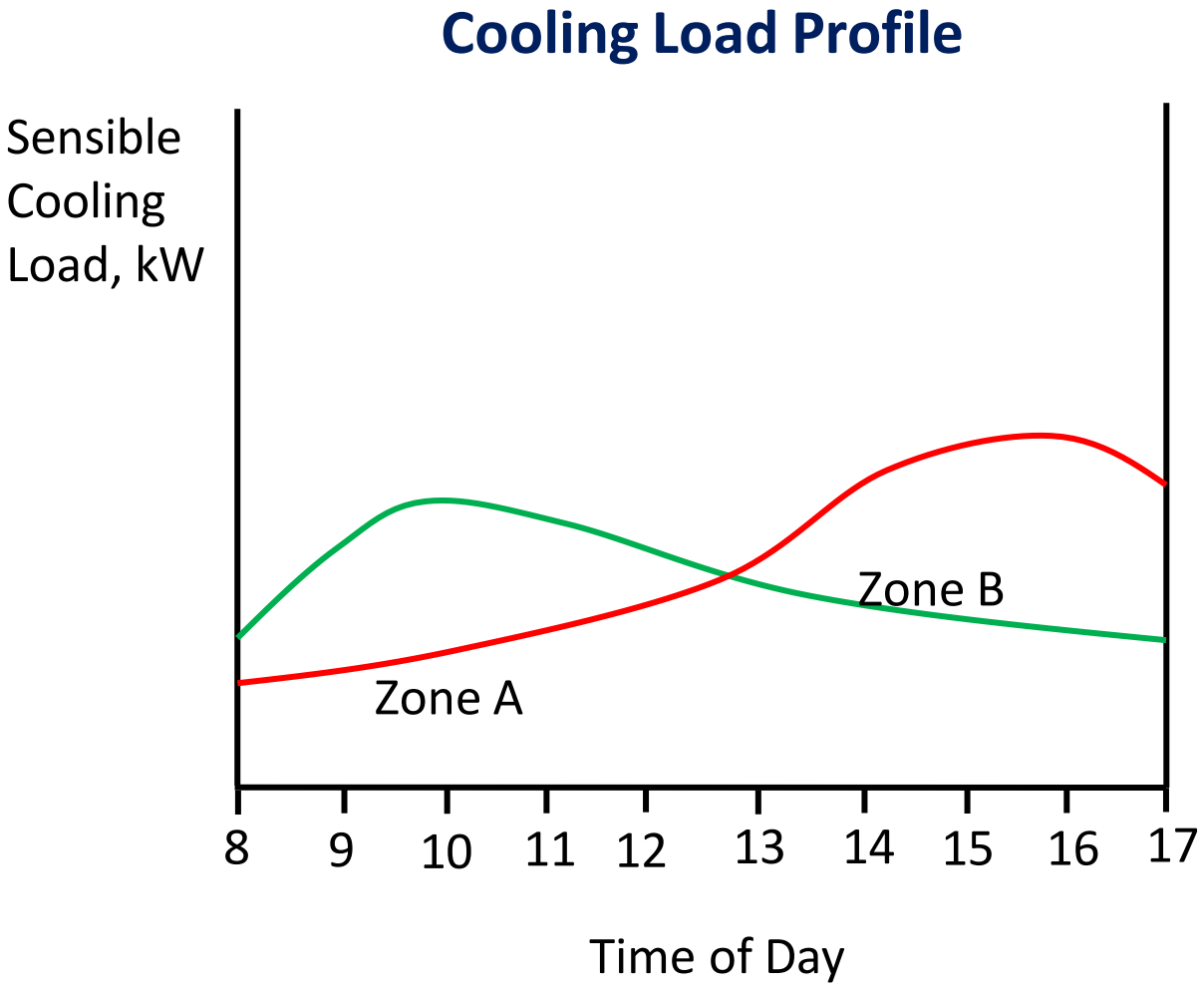
Woodlands Wafer Fab Park

District Cooling



Air Systems

- Constant Air Volume (CAV) System
- Variable Air Volume (VAV) System
- Fan Coil Systems



Air Systems

Room Sensible Cooling Load,

$RSCL = \dot{m}_s . c_{p,a} . (t_r - t_s)$

Where:

$\dot{m}_s$ = mass flow rate of supply air, kg/s

$c_{p,a}$ = specific heat of air, KJ/kgK

t_r = room temperature, °C

t_s = supply air temperature, °C

Air Systems

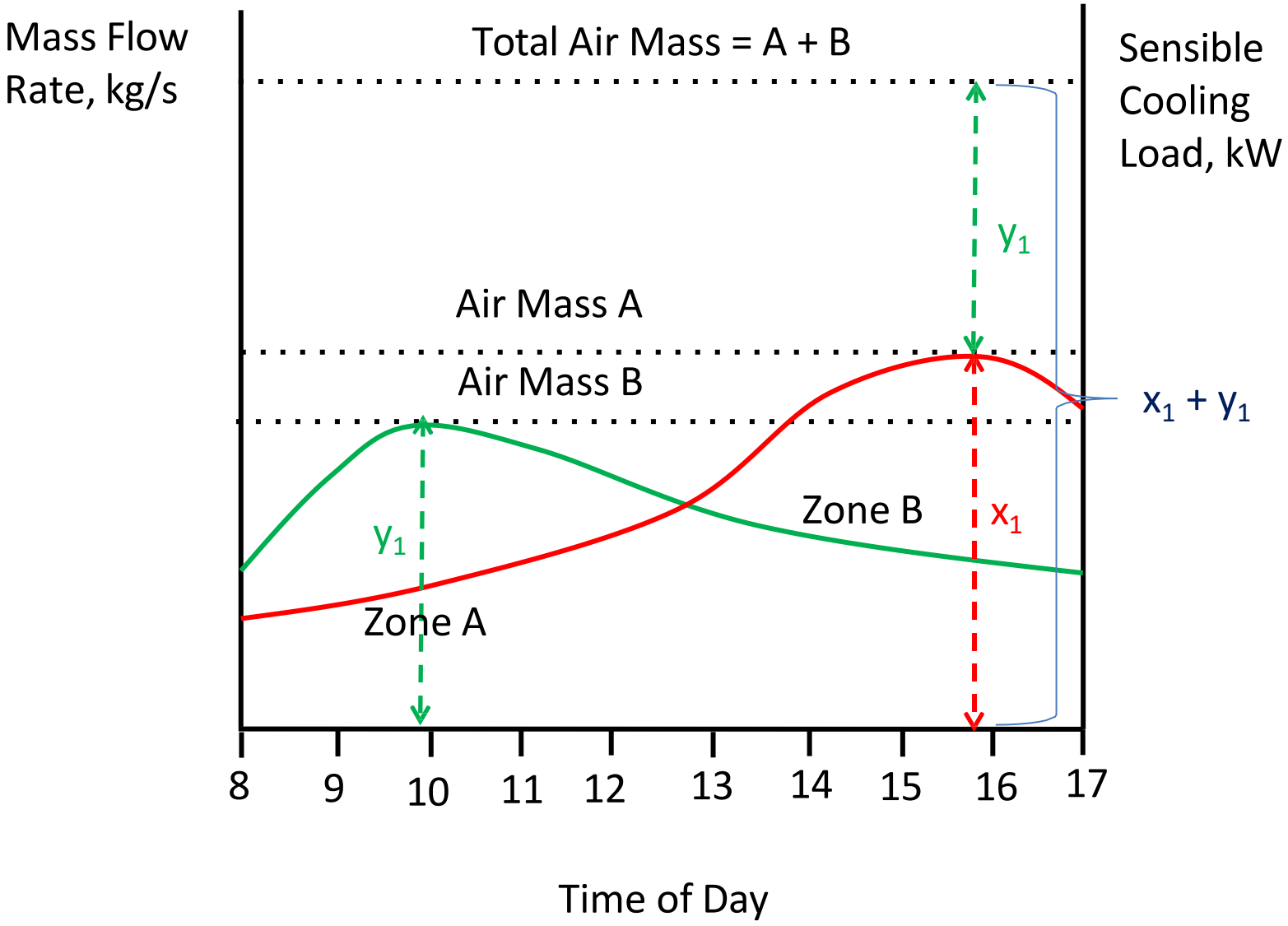
Ability of supply air to offset sensible heat gains can either be adjusted by:

- Varying supply temperature, t_s, hence $(t_r - t_s)$

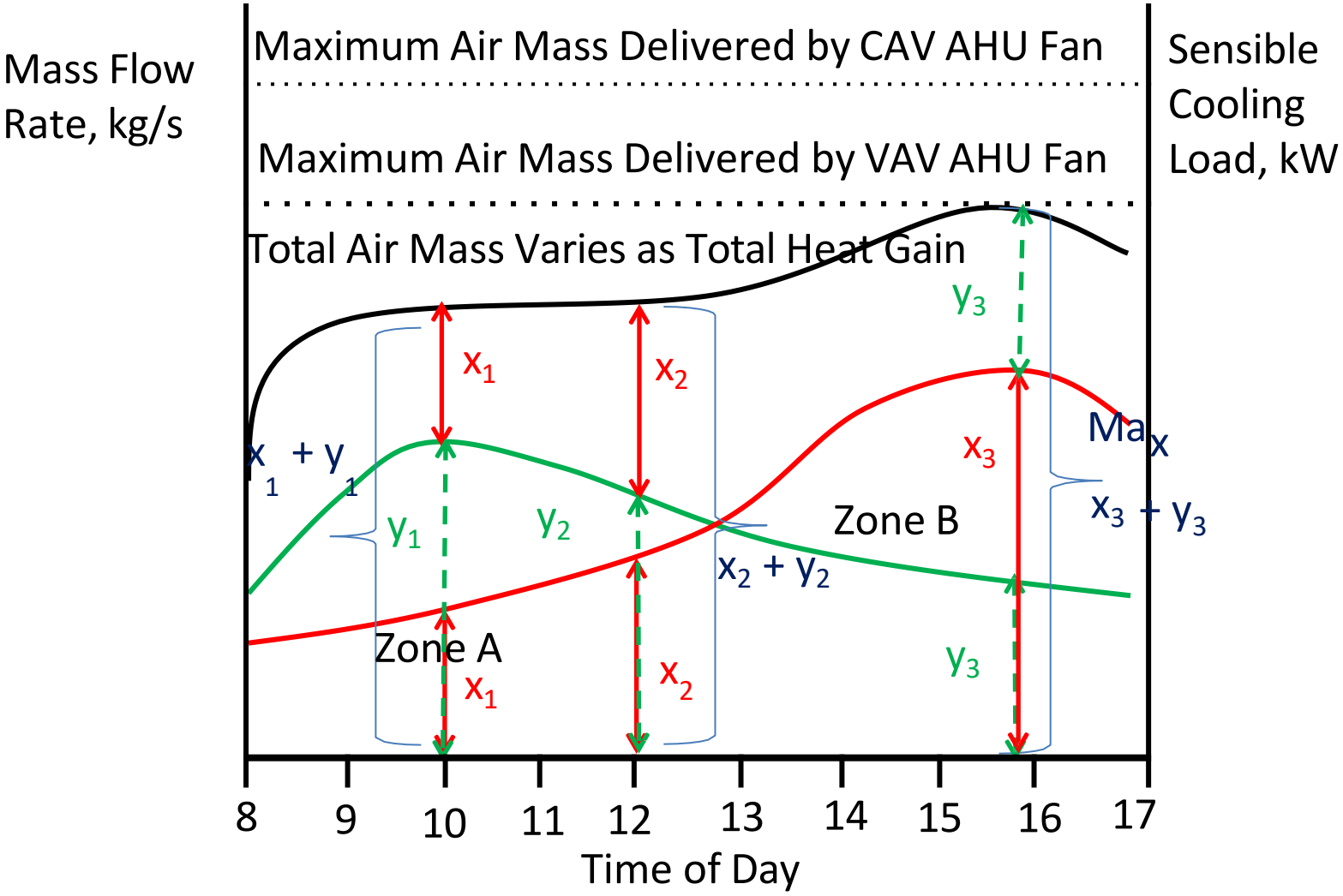
  or

- Varying mass flow rate, $\dot{m}_s$, hence the supply volume flow rate

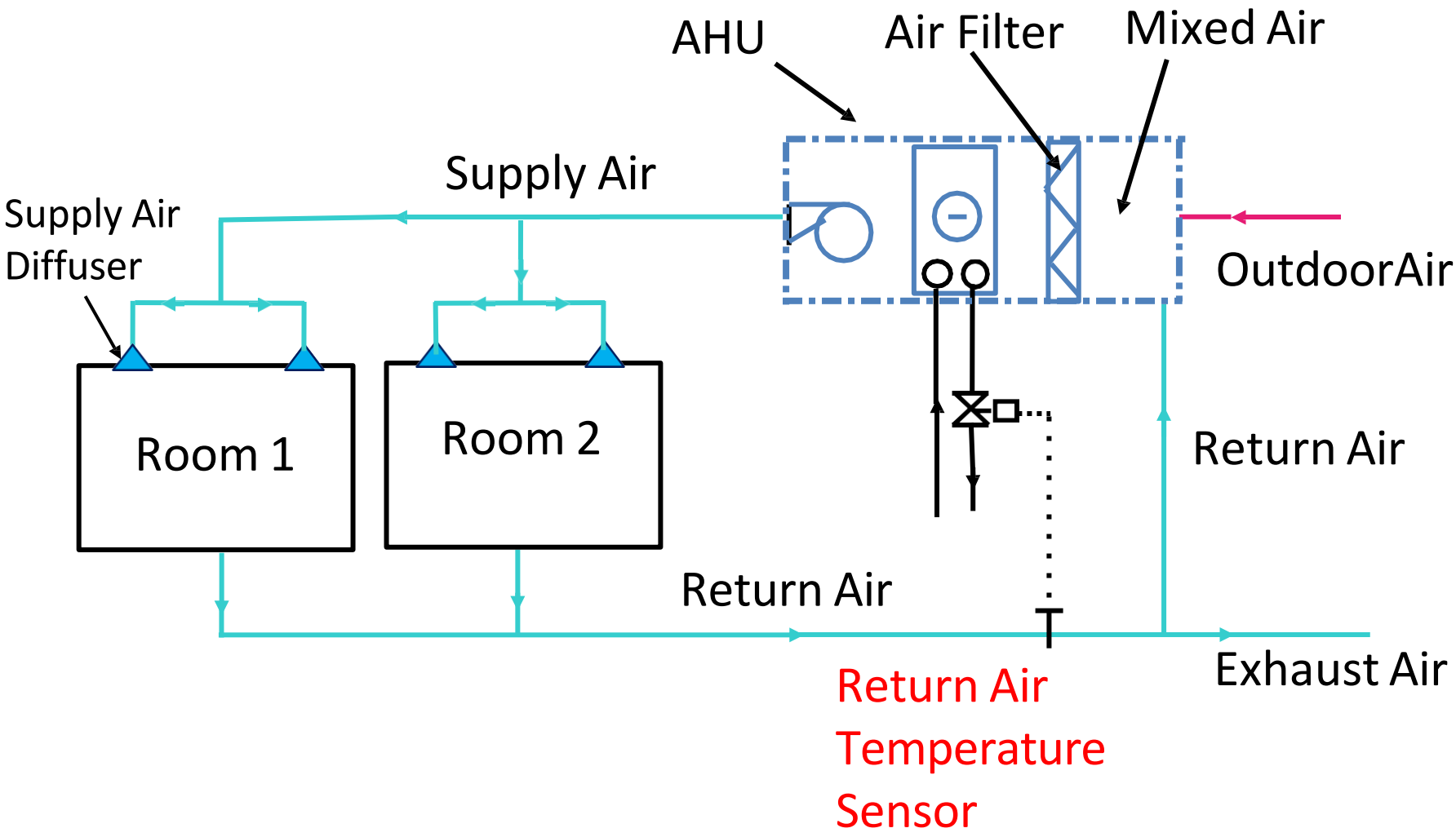
Cooling Load Profile - CAV



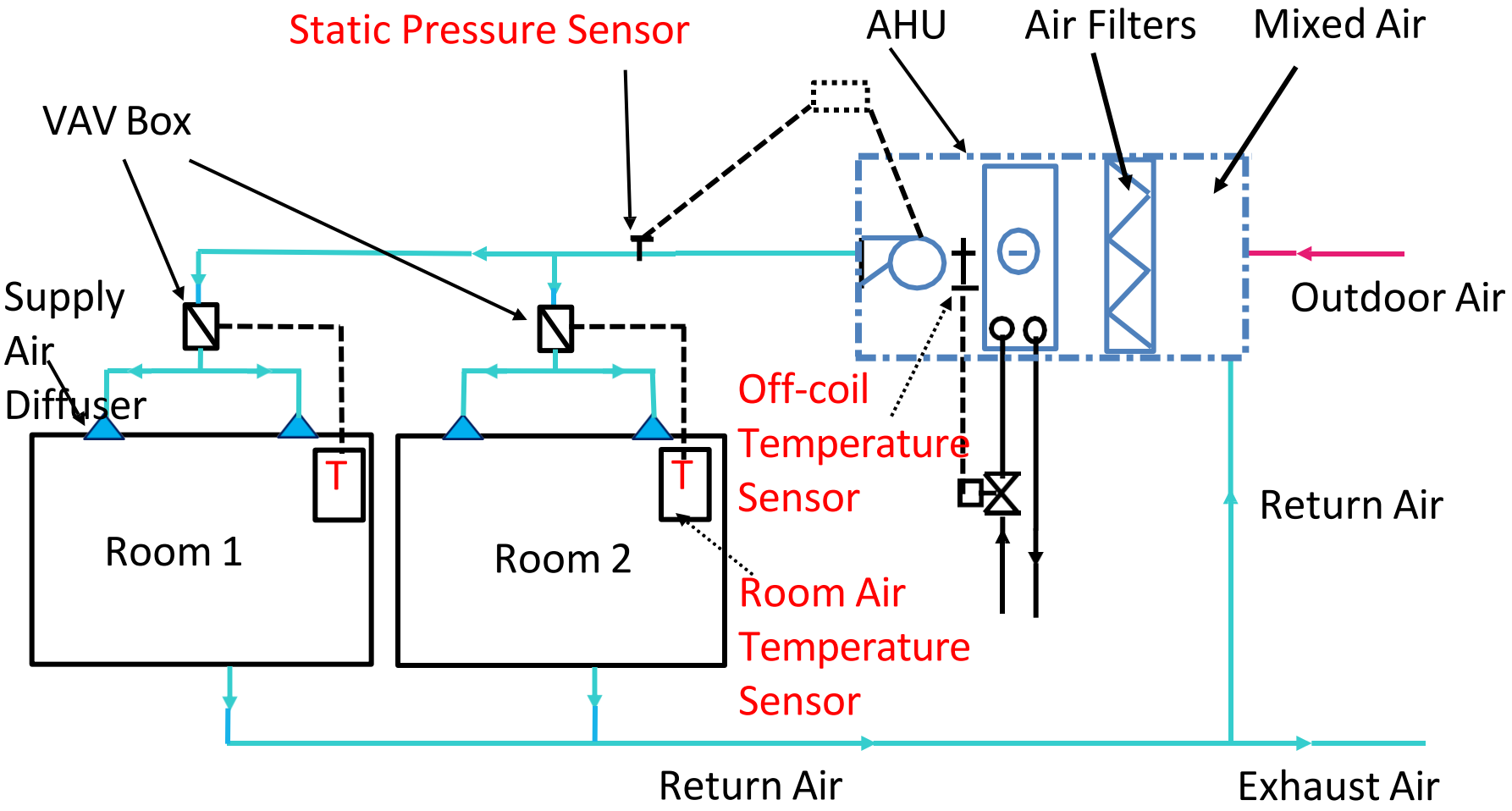
Cooling Load Profile - VAV



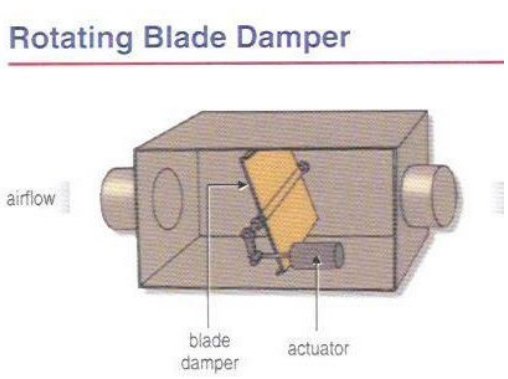
Constant Air Volume (CAV) System

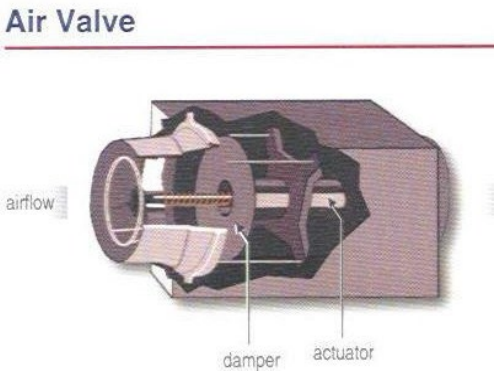


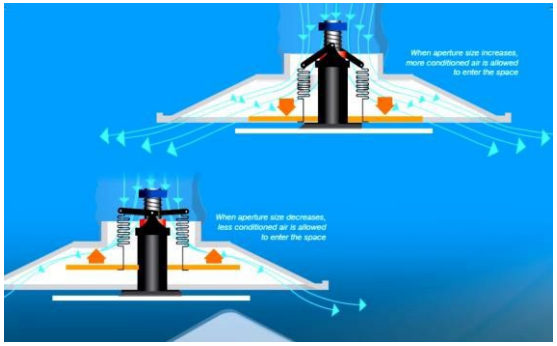
Variable Air Volume (VAV) System



Variable Air Volume (VAV) System

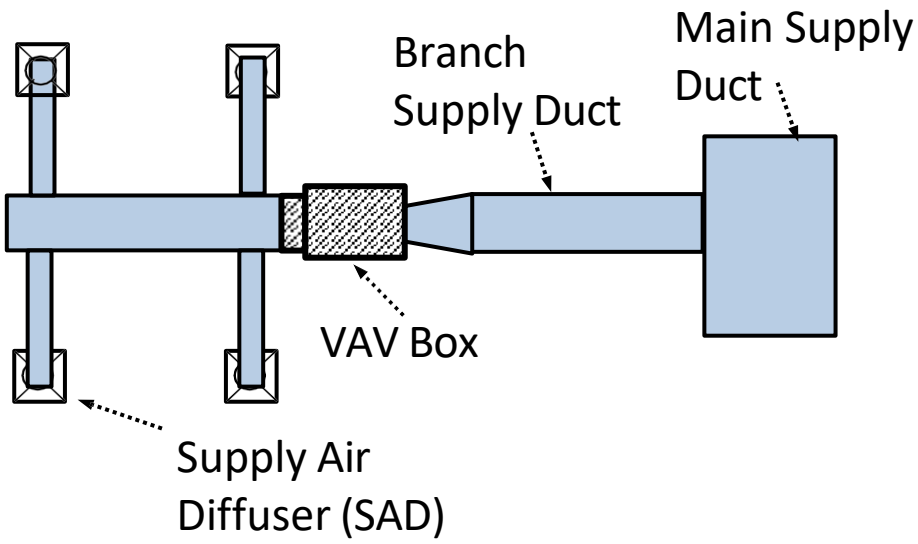


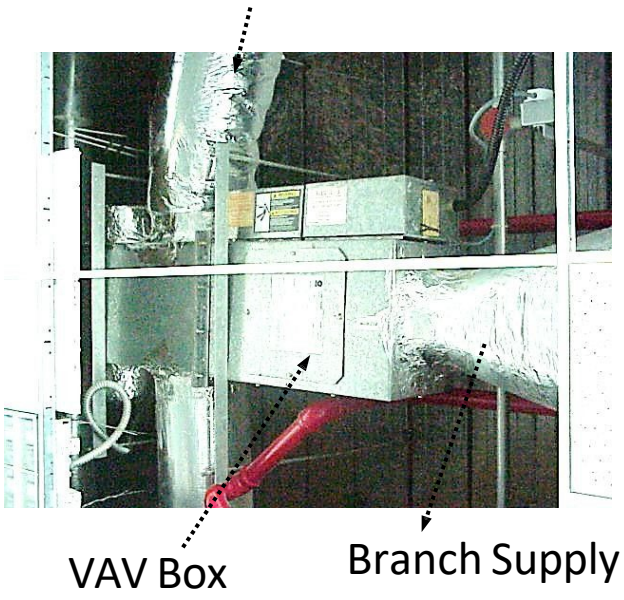




VAV Diffuser

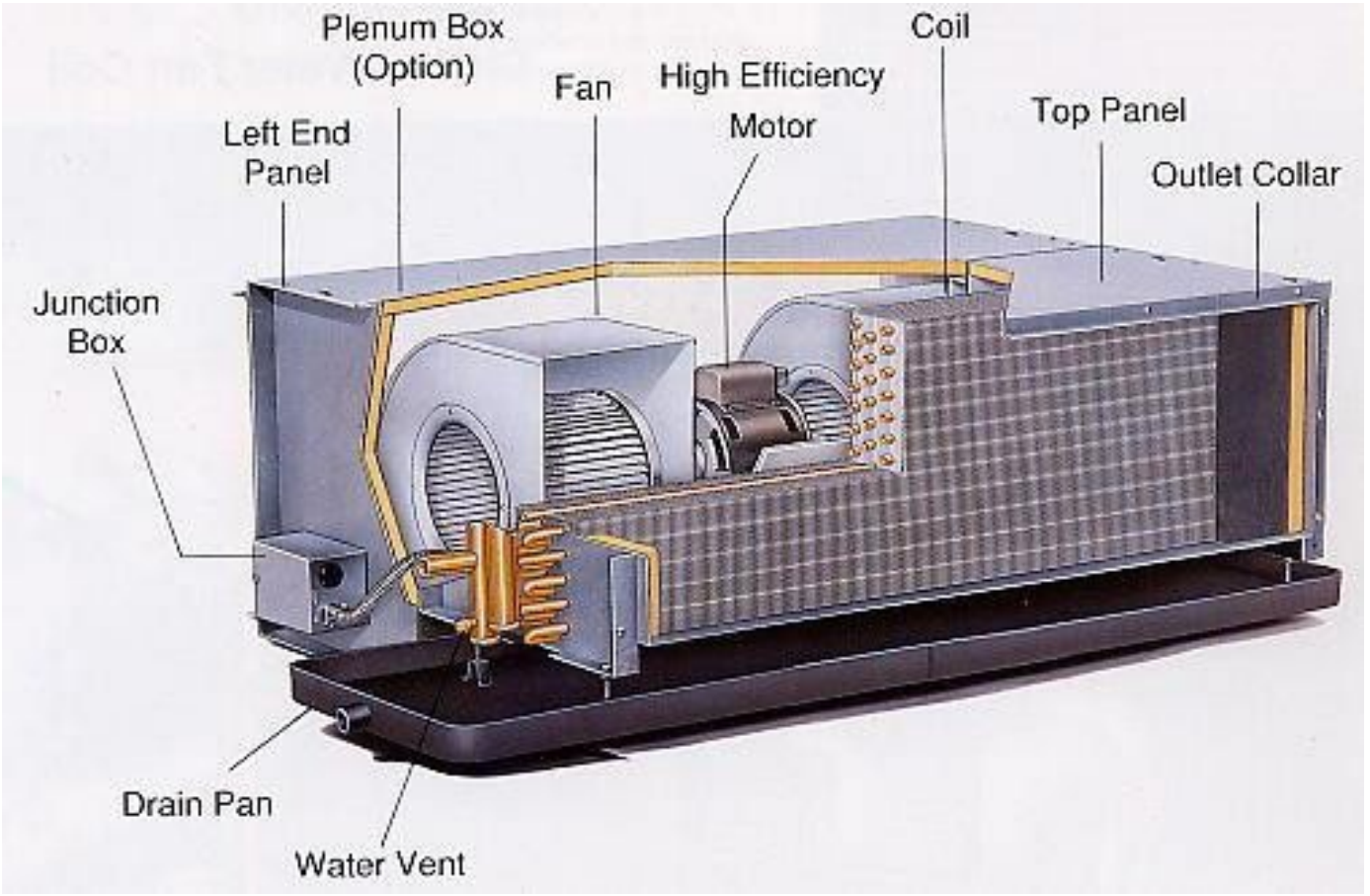
VAV Boxes without Reheat



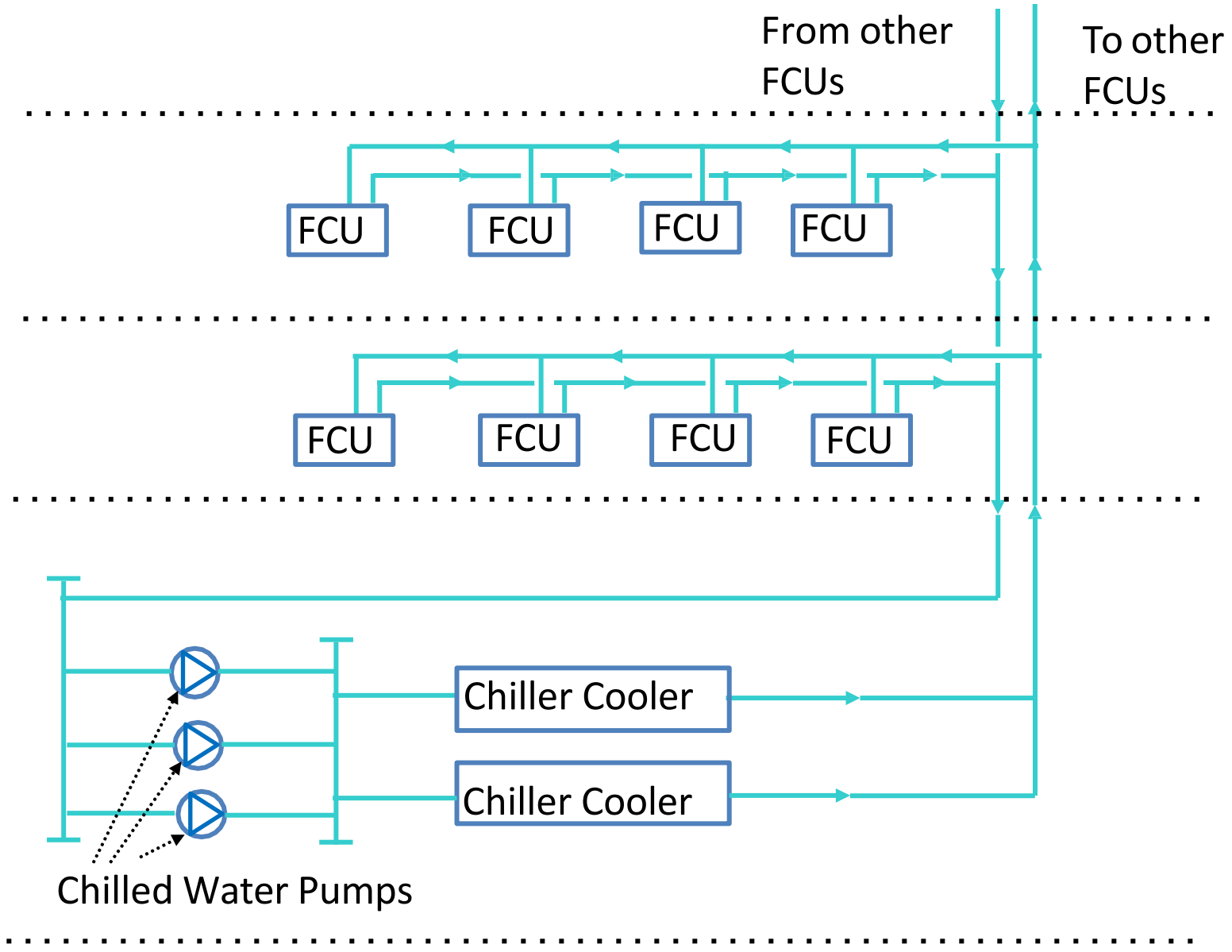


VAV Box and SAD Installation

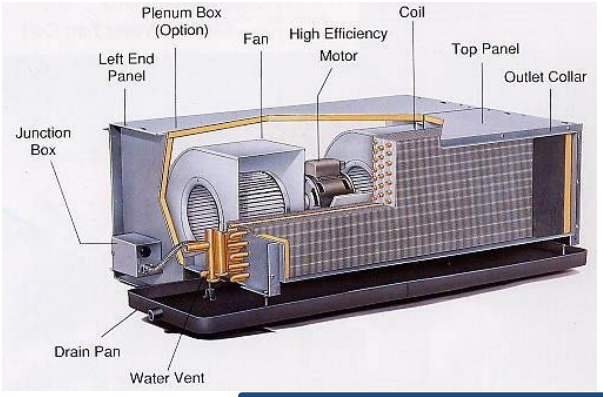
Fan Coil Unit System

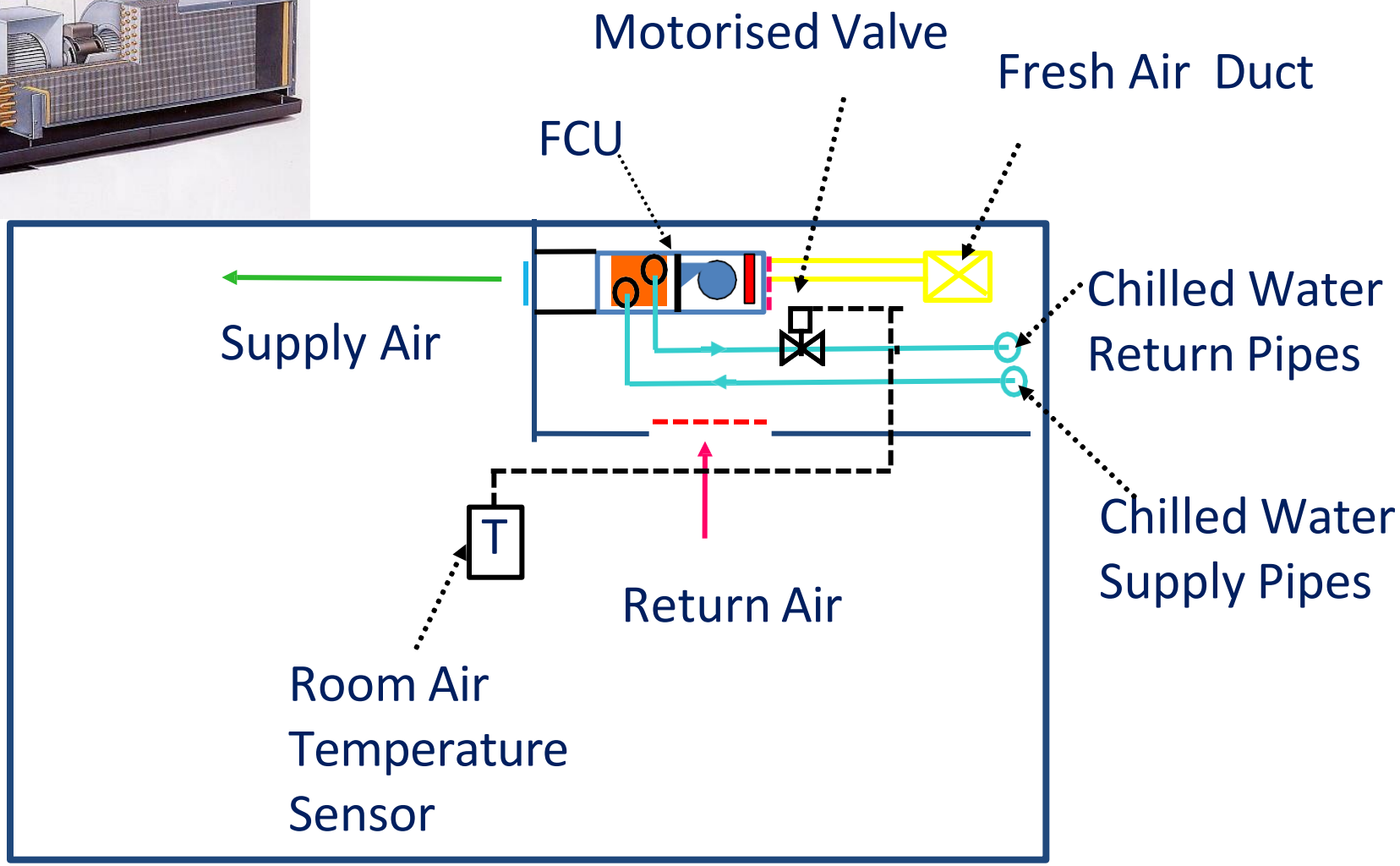


Fan Coil Unit System

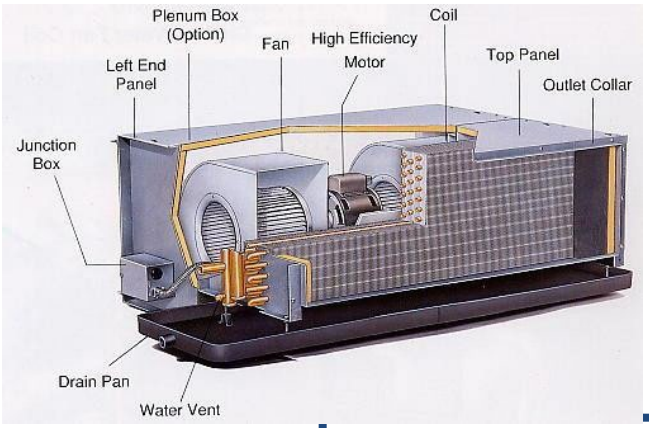


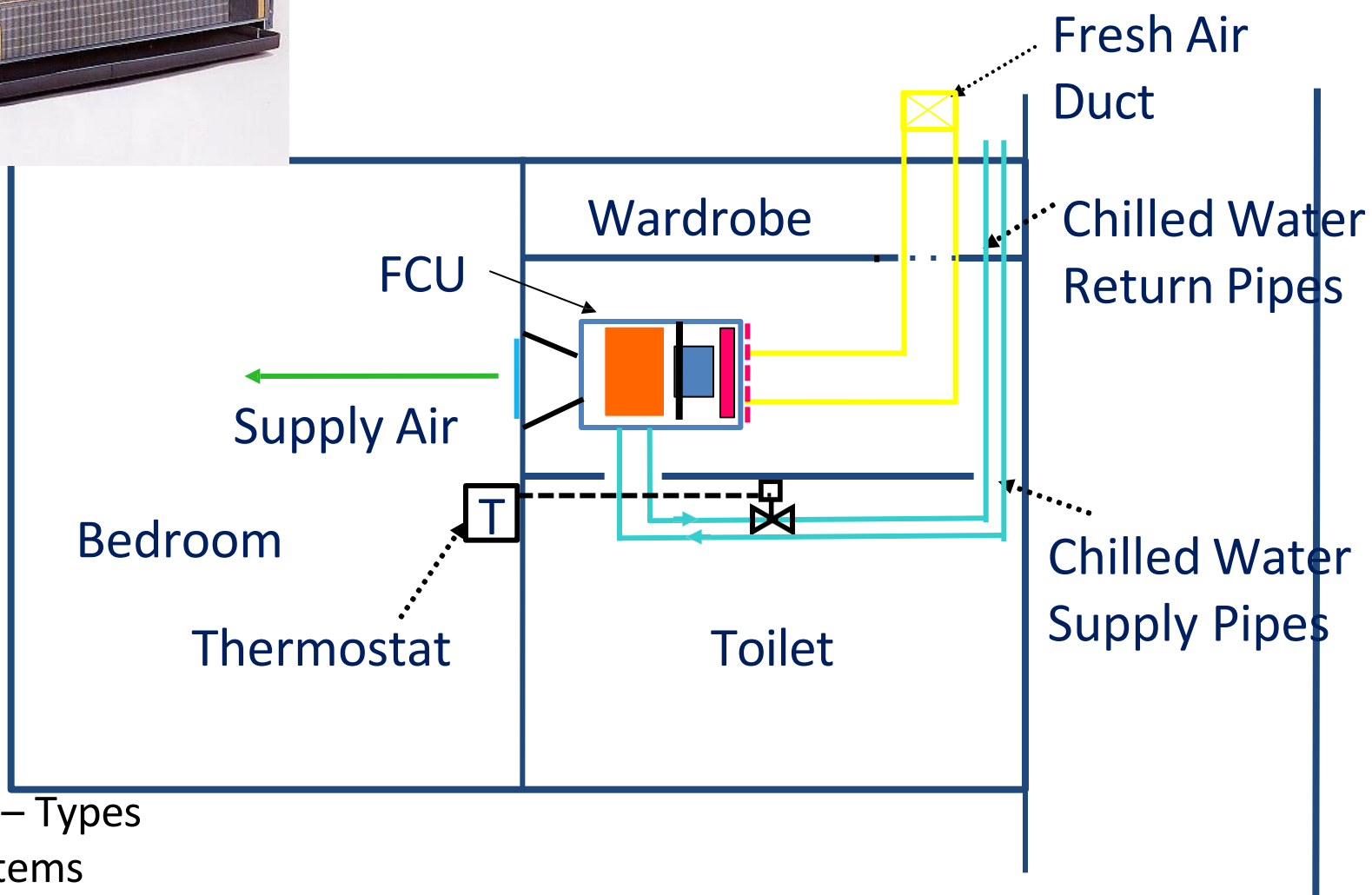
Fan Coil Unit System





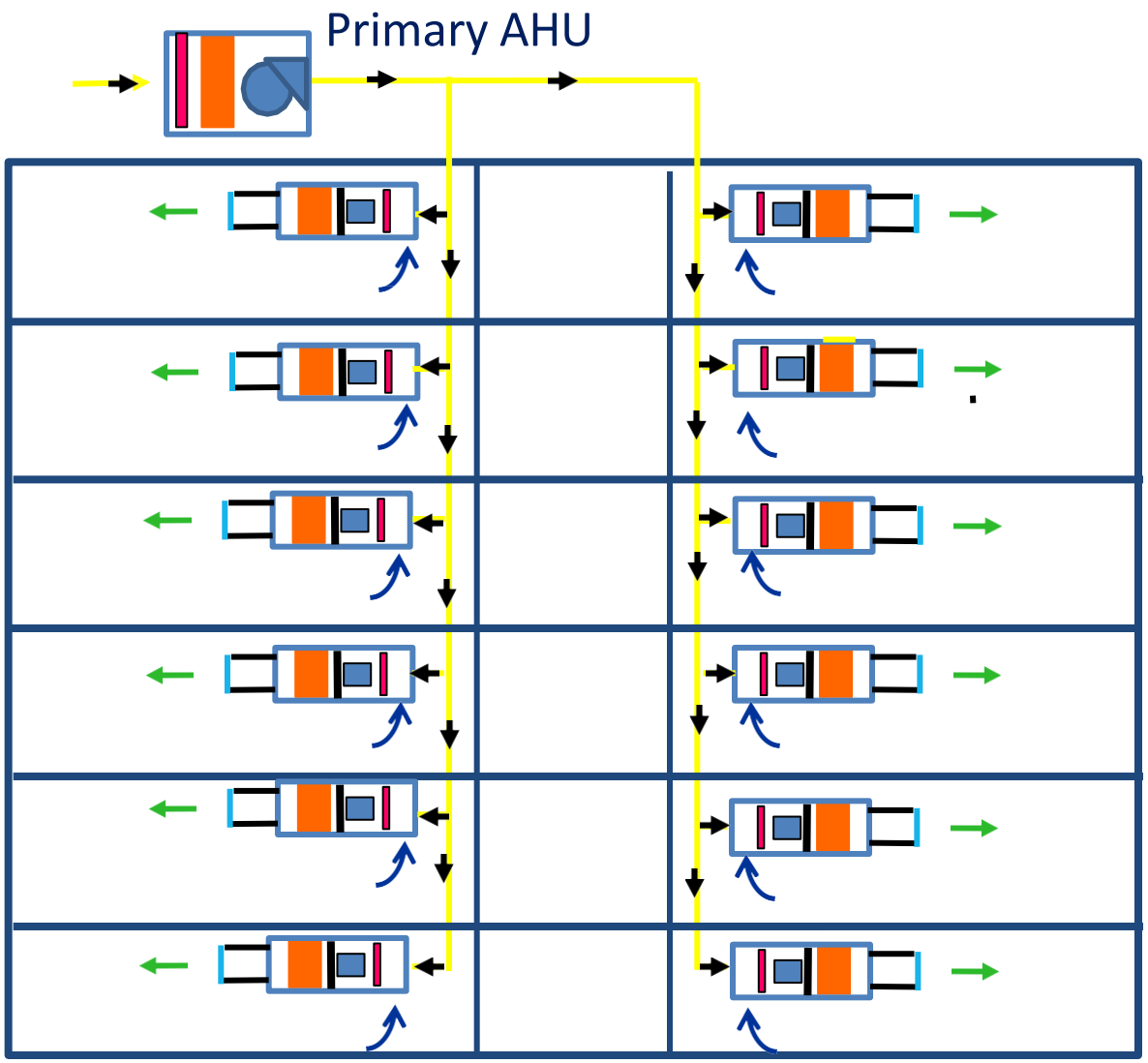
Fan Coil Unit System



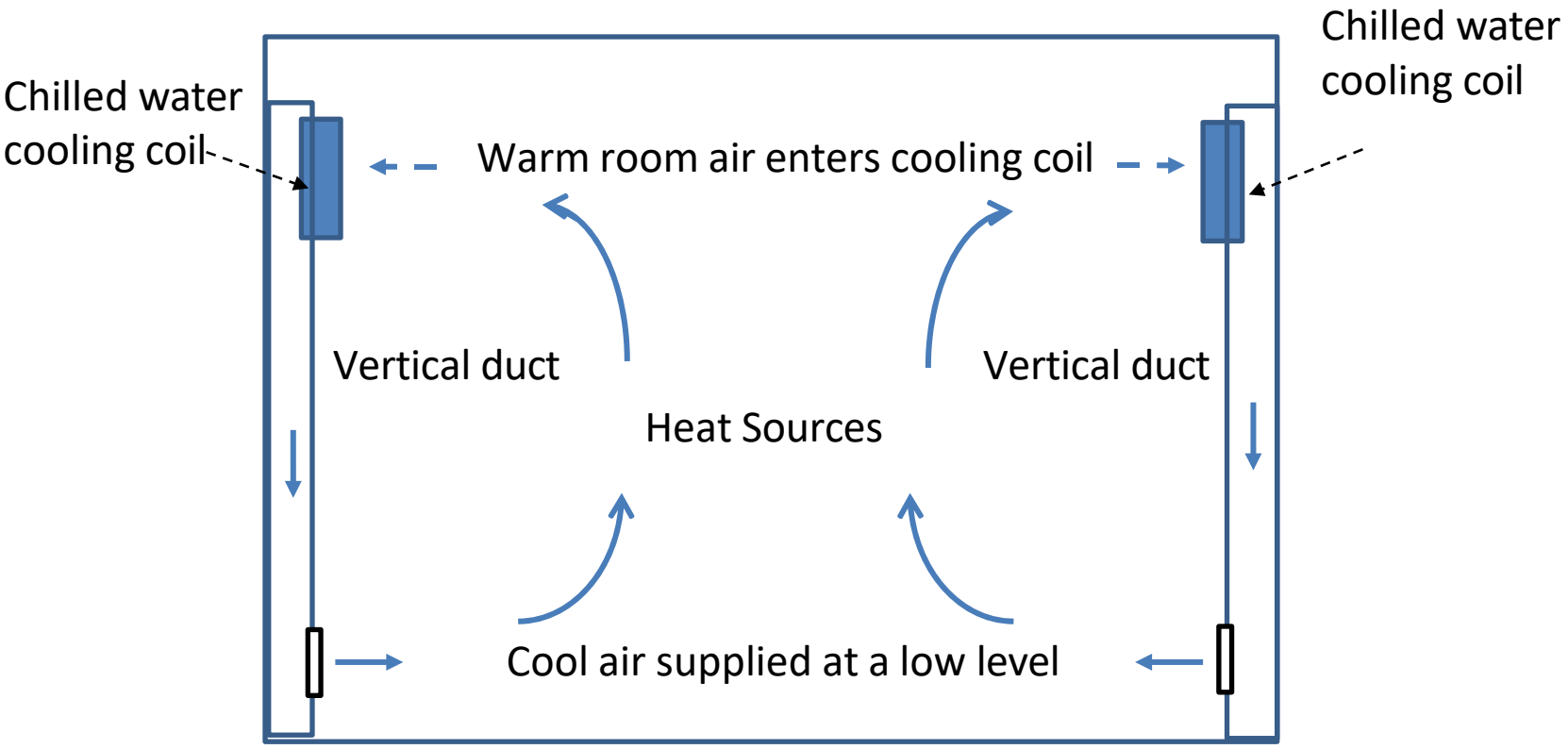


Video – Types of Systems

Fan Coil Unit System



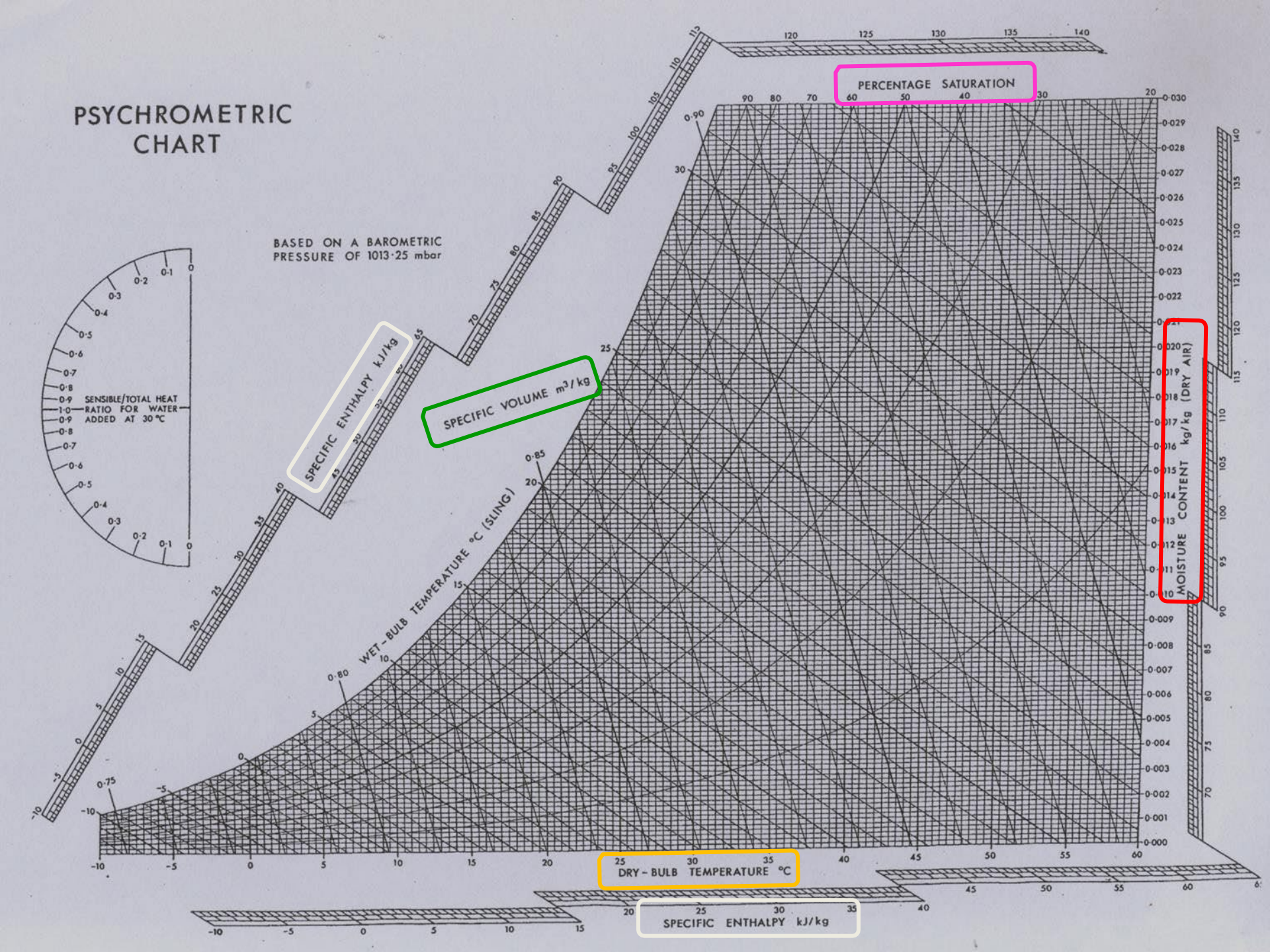
Passive Displacement Cooling System



Applied Psychrometrics

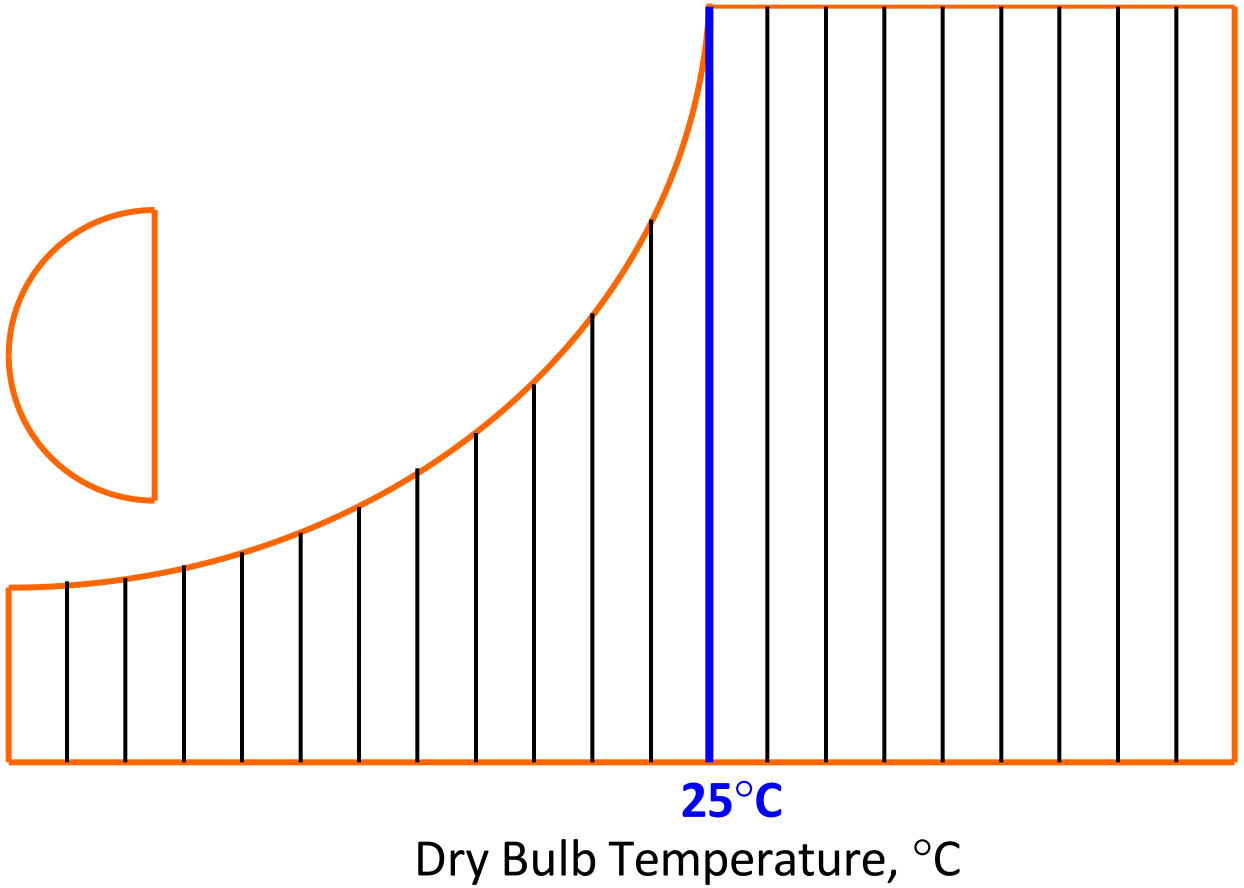
Properties of moist air which are of interest in air-conditioning are:

- Dry bulb temperature
- Wet bulb temperature
- Enthalpy
- Moisture content
- Relative Humidity (~ % saturation)
- Density or specific volume


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30 °C
SPECIFIC ENTHALPY kJ/kg
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE °C (SLING)
PERCENTAGE SATURATION
MOISTURE CONTENT kg/kg (DRY AIR)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg

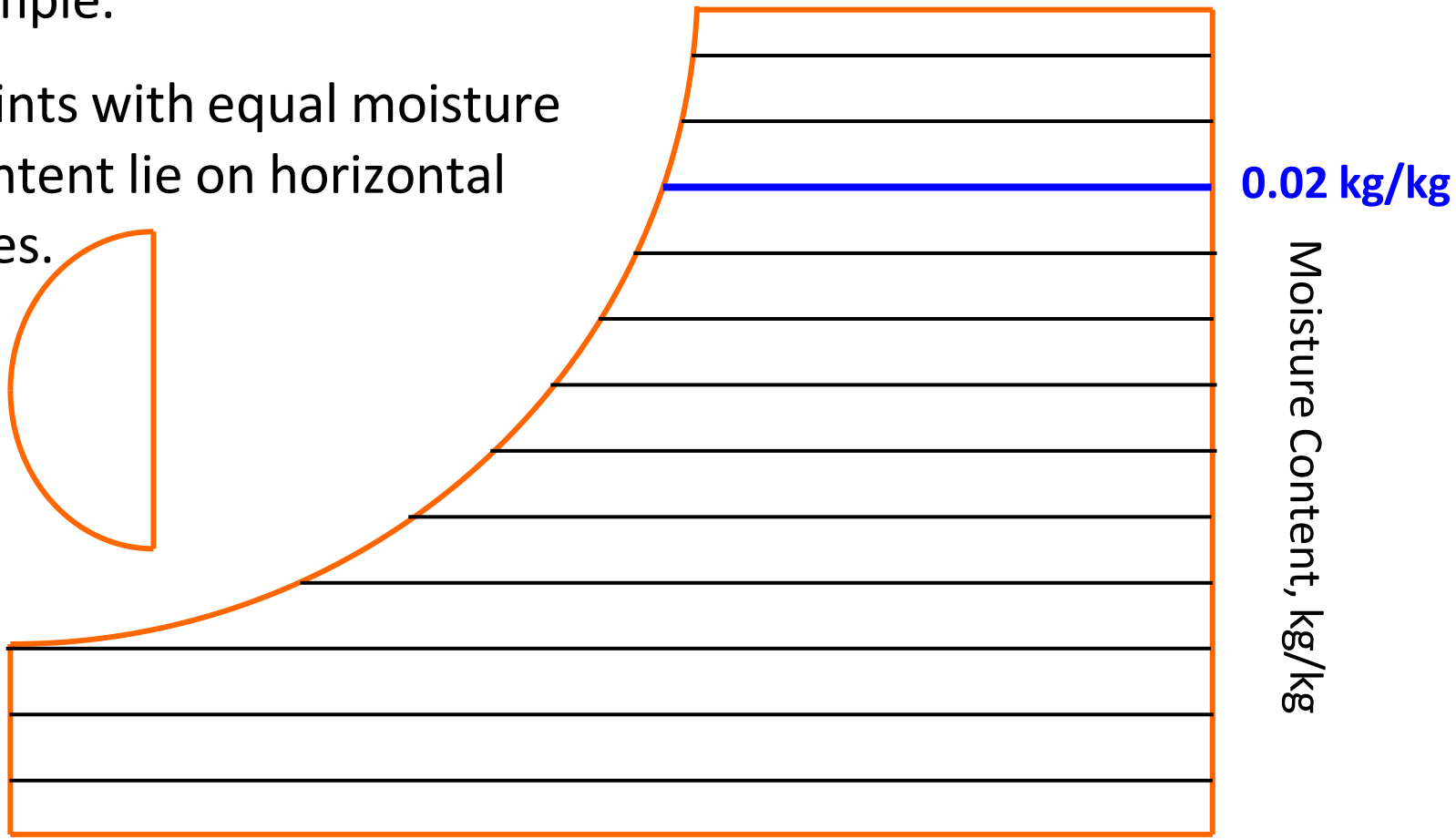
Dry Bulb Temperature

- Points at constant dry bulb temperatures lie on vertical lines.



Moisture Content

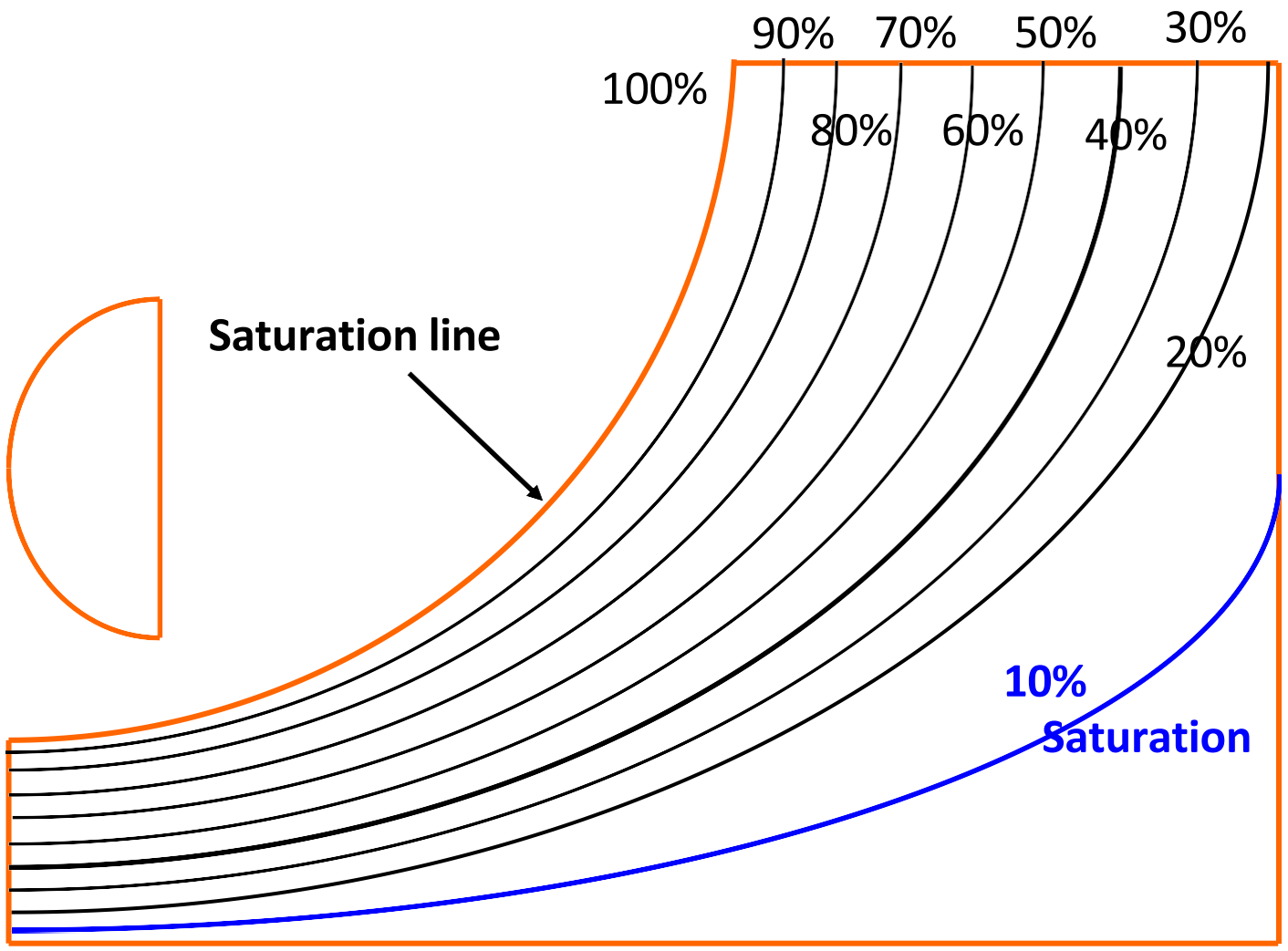
- Ratio of mass of moisture (water vapour) in air to total mass of dry air in sample.
- Points with equal moisture content lie on horizontal lines.



% Saturation

- Amount of moisture in air compared to amount of moisture when it is saturated, at the same temperature.
- When air is saturated with moisture, % saturation (and RH) is 100%.

% Saturation



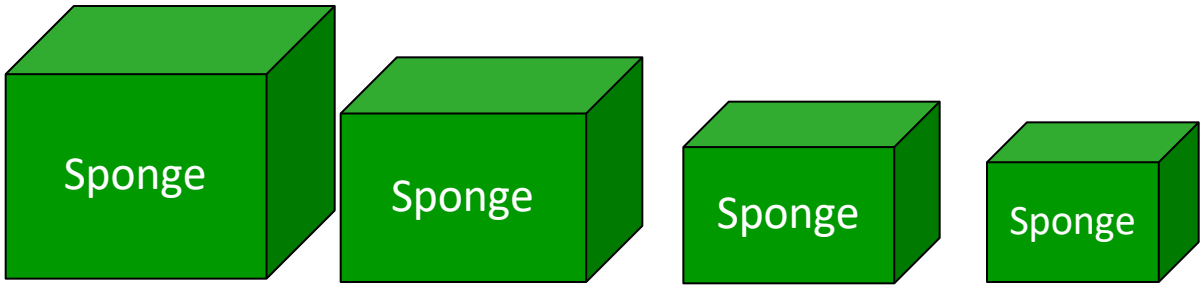
Dew Point

- Temperature at which moisture in air would start to condense.
- Dry bulb temperatures on saturation line.
- Dry bulb temperature = wet bulb temperature

Dew Point

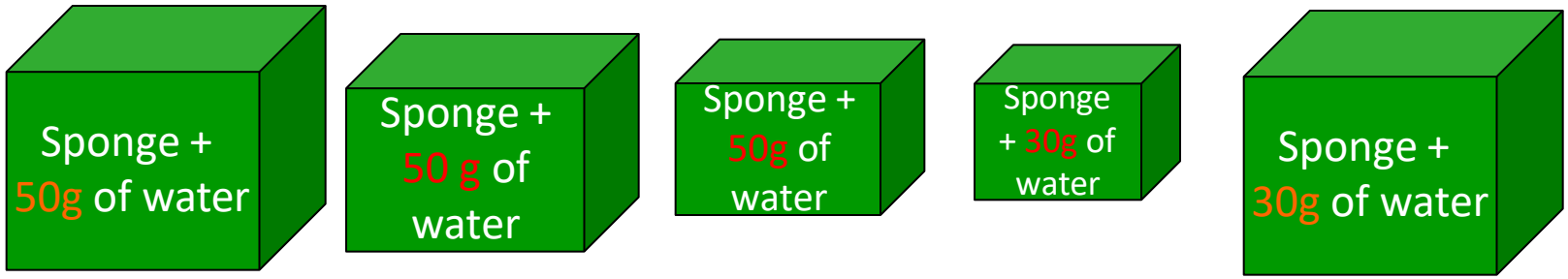
Sponge ≡ Air Lower temp ≡ Smaller sponge

25°C 22°C 19°C 16°C



Max amount of water it can hold is

100 g 80 g 50 g 30 g



RH = $\frac{50}{100}$ * 100% = 50%

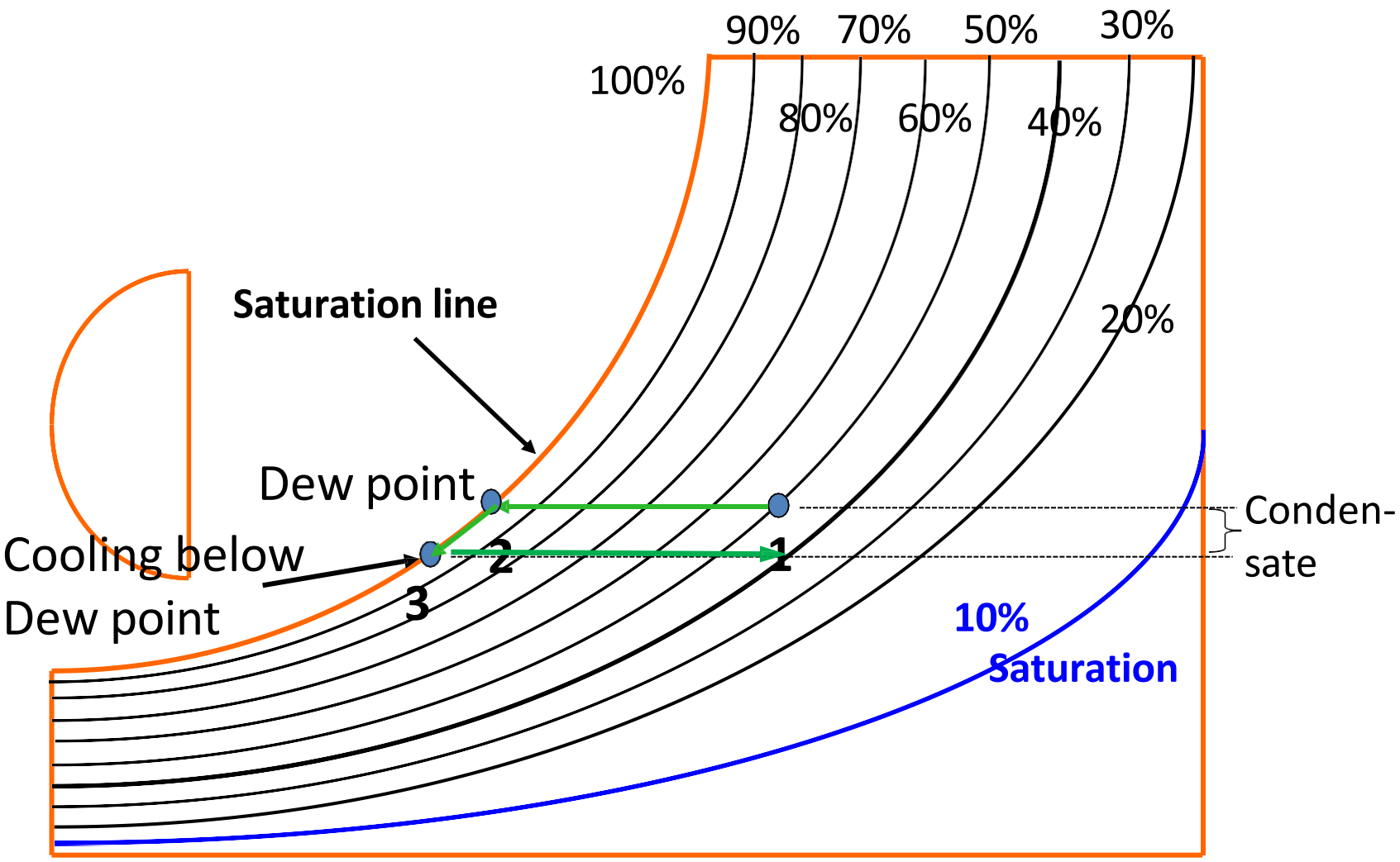
RH = 62.5%

RH = 100%

RH = 100% + 20g of condensate

RH = 30%

Dew Point


90%
70%
50%
30%
100%
80%
60%
40%
20%
Saturation line
Dew point
Cooling below
Dew point
2
1
3
Conden-
sate
10%
Saturation

Wet Bulb Temperature

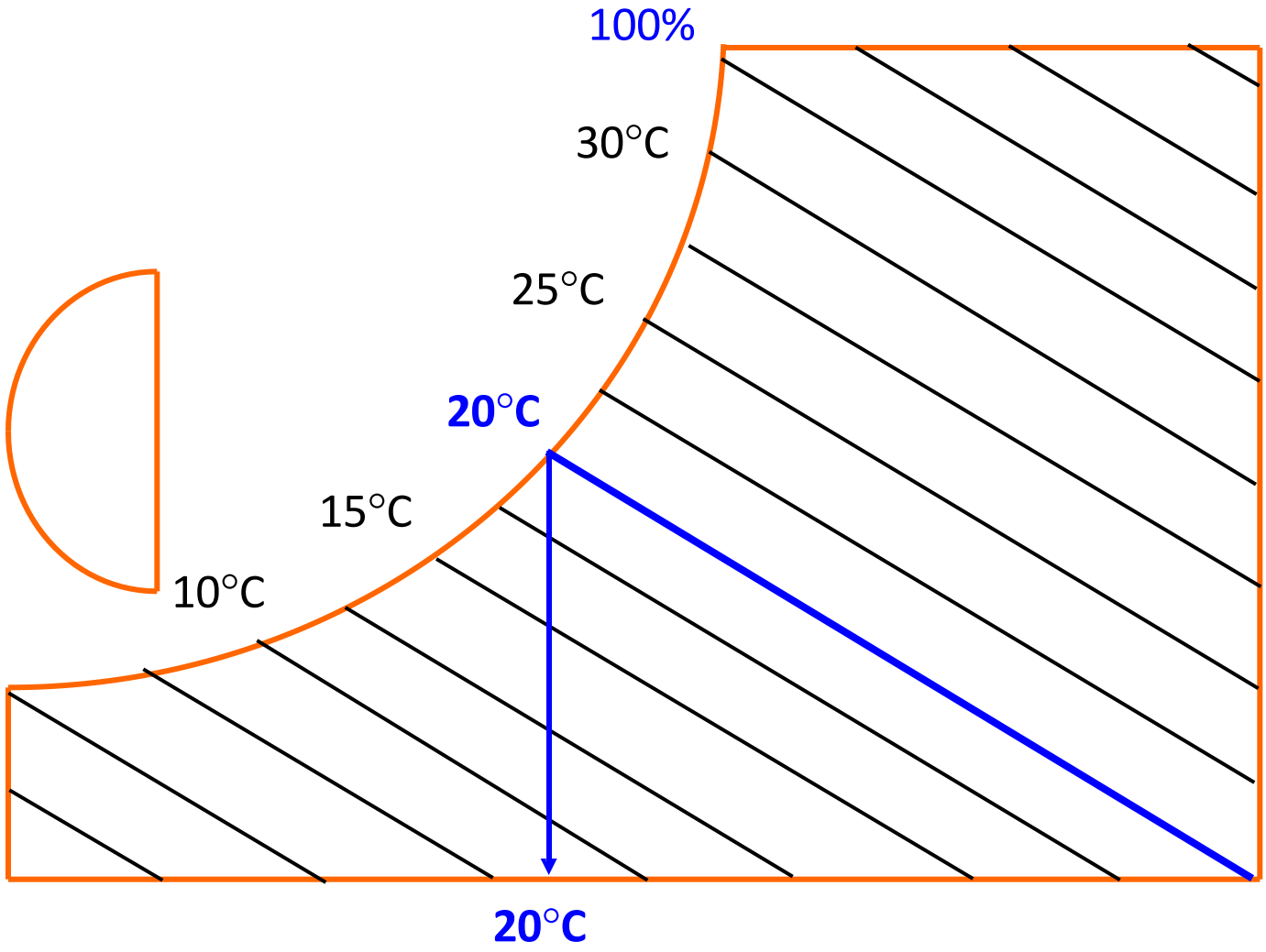
- Temperature indicated when bulb of thermometer is kept moist.
- Equal or lower than that of dry bulb thermometer.
- % saturation low, wet bulb temperature considerably lower than dry bulb temperature.



Wet Bulb Temperature

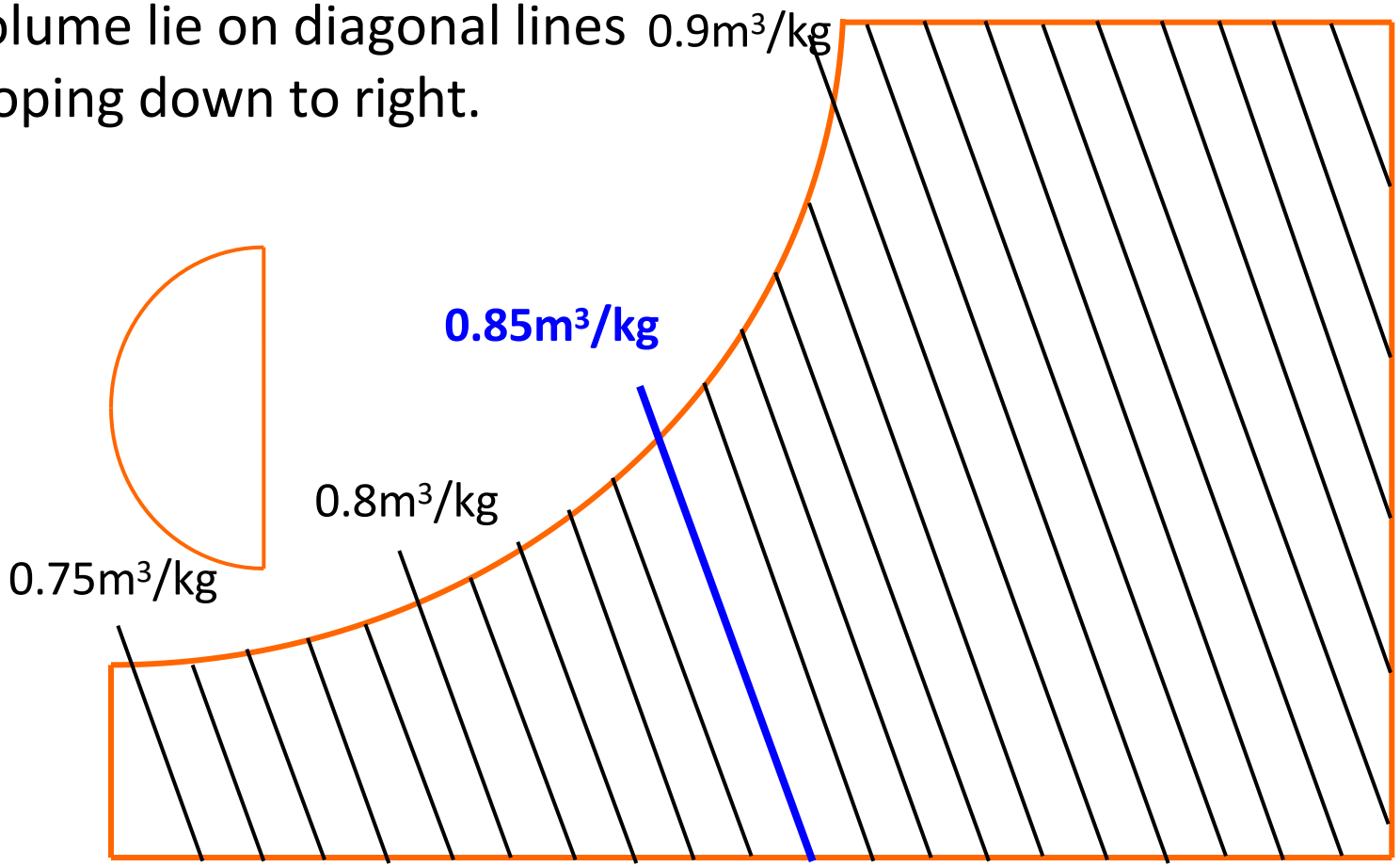
- % saturation high, dry and wet bulb temperatures would be quite close.
- At 100% saturation, wet bulb and dry bulb temperatures are the same.
- Points having same wet bulb temperature lie on diagonal lines sloping down to the right.

Wet Bulb Temperature



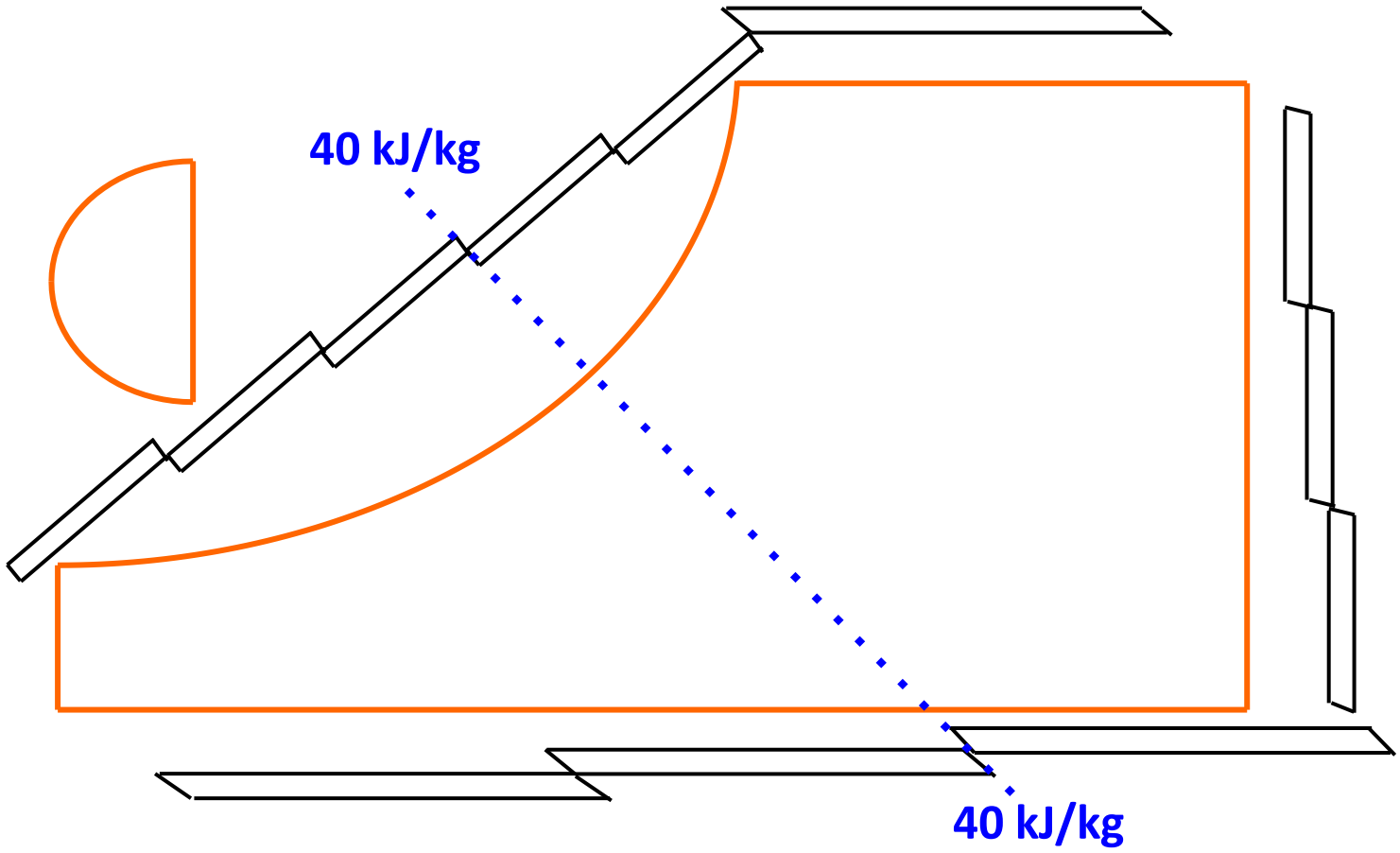
Specific Volume

- Points having same specific volume lie on diagonal lines sloping down to right.



Enthalpy

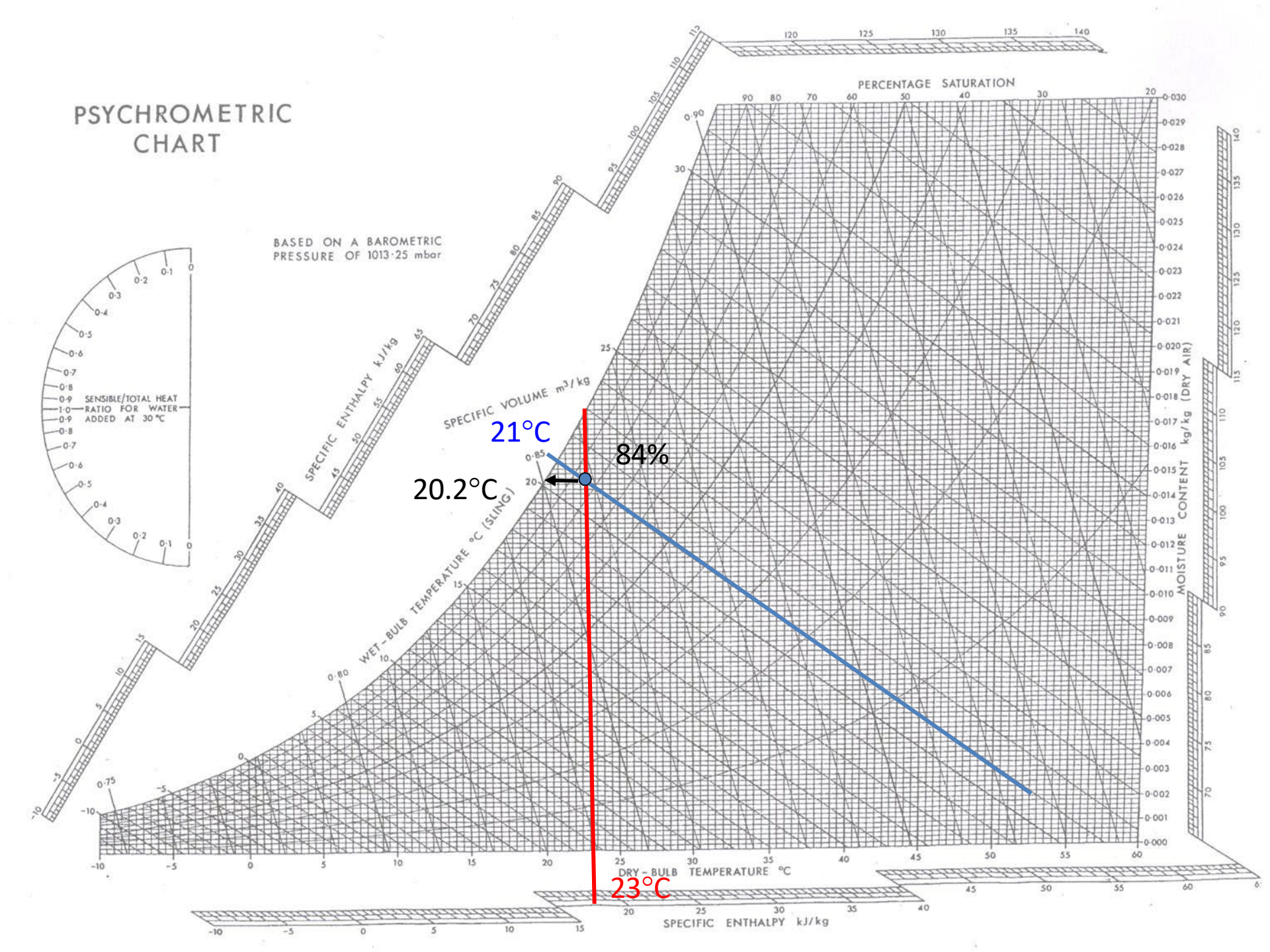
- Heat content of moist air.

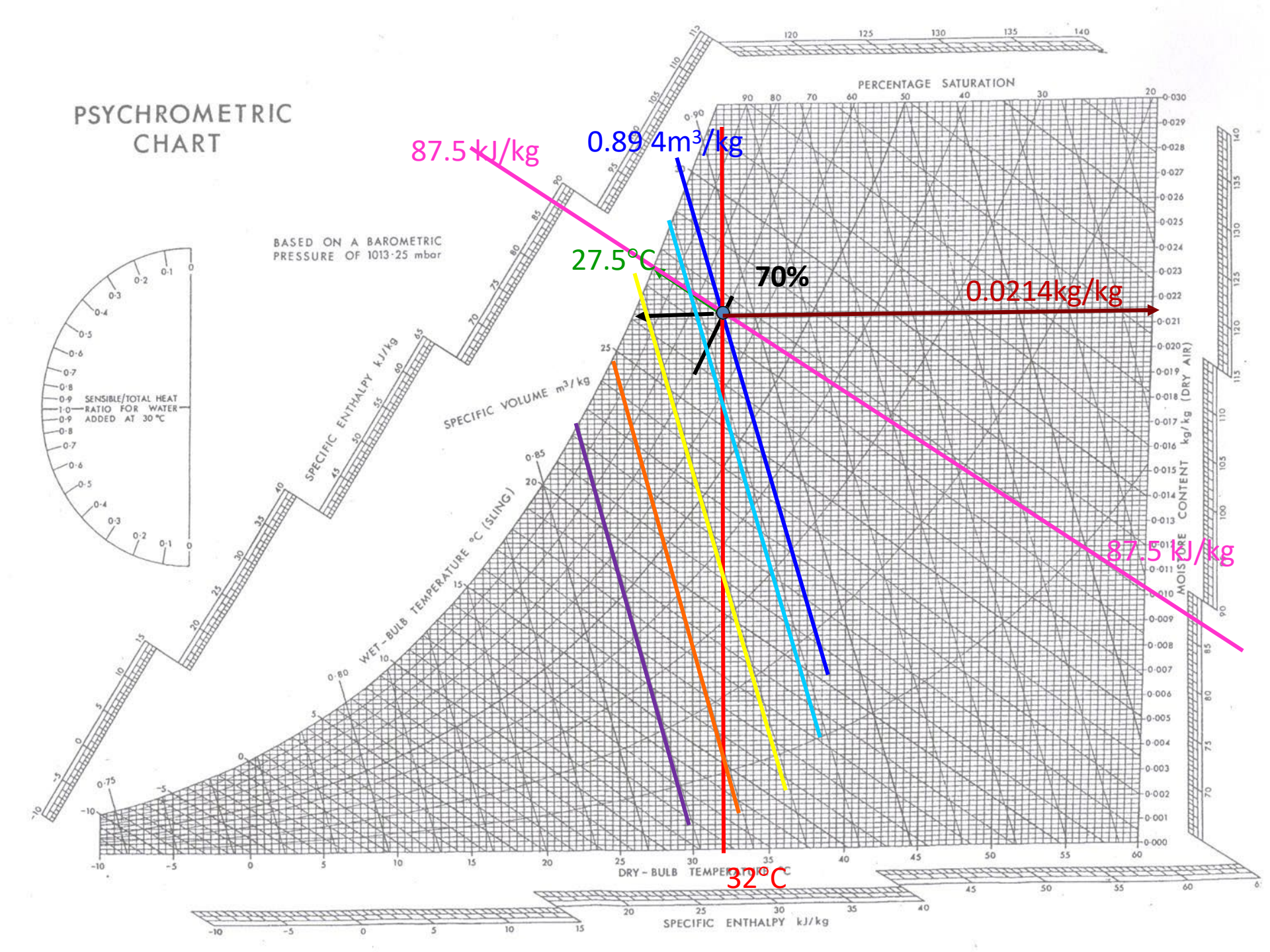


Example 1.7

Use the psychrometric chart to determine the following:

- Room air condition measured at 23°C dry bulb temperature and 21°C wet bulb temperature. What is the % saturation and dew point?
  Ans: 84%, 20.2°C
- Outside air condition measured at 32°C dry bulb temperature, 70% saturation. Determine the wet bulb temperature, specific volume, moisture content and enthalpy.
  Ans: 27.5°C wb; 0.894m^3/kg;0.0214kg/kg; 87.5kJ/kg


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30 °C
PERCENTAGE SATURATION
SPECIFIC VOLUME m³/kg
SPECIFIC ENTHALPY kJ/kg
WET-BULB TEMPERATURE °C (SLING)
DRY-BULB TEMPERATURE °C
MOISTURE CONTENT kg/kg (DRY AIR)
SPECIFIC ENTHALPY kJ/kg
21°C
84%
20.2°C
23°C


PSYCHROMETRIC CHART
BASED ON A BAROMETRIC PRESSURE OF 1013·25 mbar
87.5 kJ/kg
0.894m³/kg
27.5°C
70%
0.0214kg/kg
87.5 kJ/kg
32°C
PERCENTAGE SATURATION
SPECIFIC VOLUME m³/kg
WET-BULB TEMPERATURE °C (SLING)
DRY-BULB TEMPERATURE °C
SPECIFIC ENTHALPY kJ/kg
MOISTURE CONTENT kg/kg (DRY AIR)
SENSIBLE/TOTAL HEAT RATIO FOR WATER ADDED AT 30 °C

Basic Air-conditioning Processes

Air-conditioning is the automatic control of an artificial environment within a building enclosure for:

- Comfort of human beings
- Proper performance of some industrial or scientific process
- Maintaining the quality and life of a stored product.

Basic Air-conditioning Processes

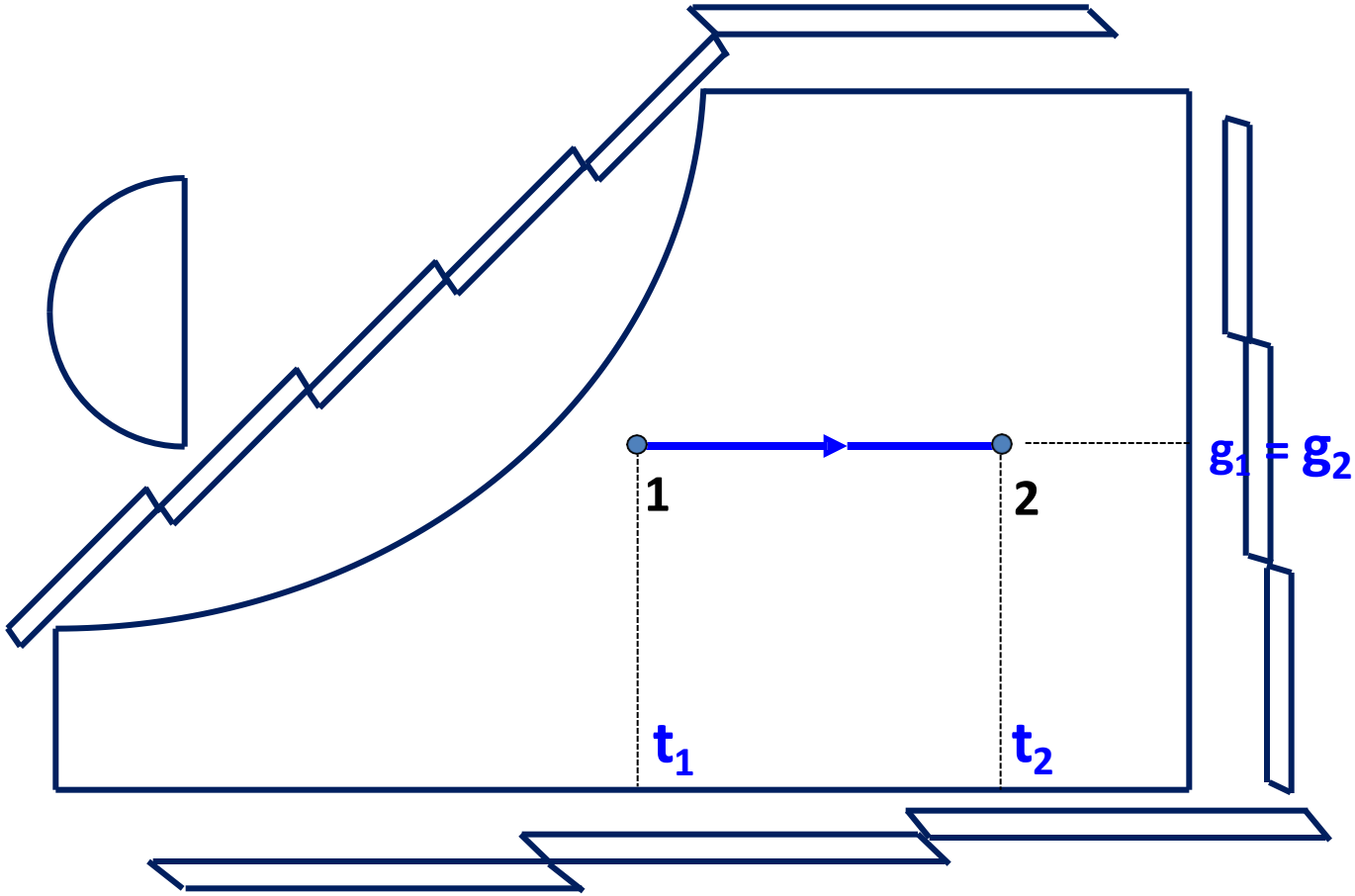
Artificial environment is created by simultaneously controlling:

- Temperature
- Relative Humidity (Moisture content)
- Air Movement
- Cleanliness
- Air Quality
- Mean Radiant Temperature (not in local context)

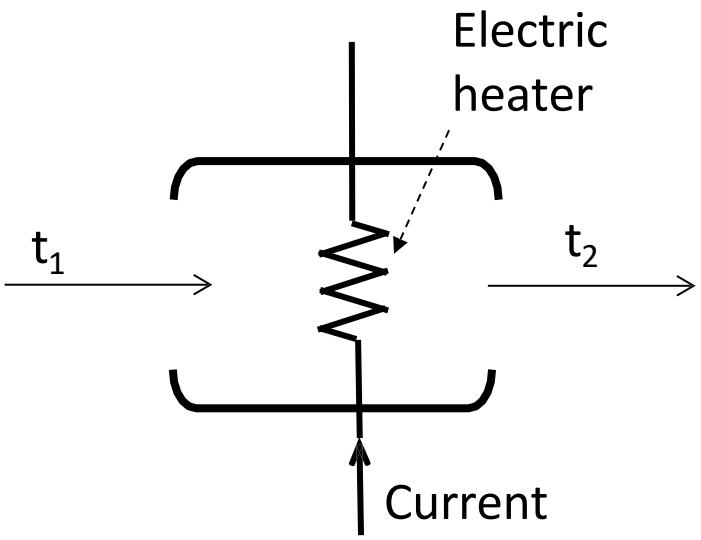
Major Air-conditioning Processes

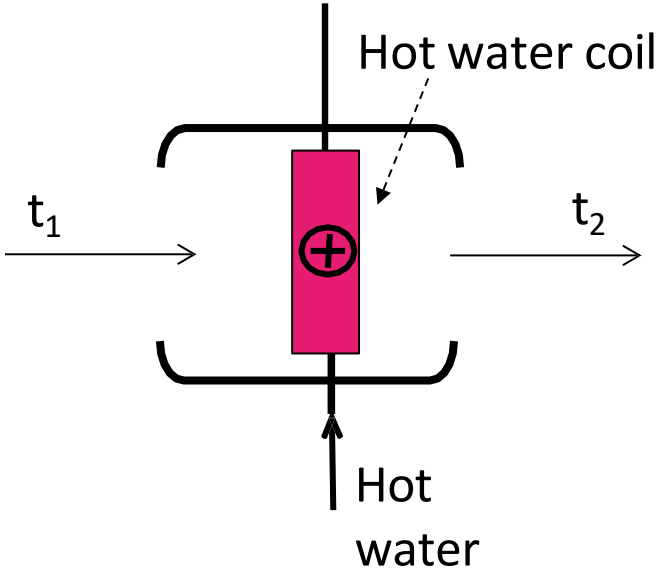
- Heating
- Cooling
- Humidifying
- Dehumidifying
- Cleaning
- Ventilating
- Air Movement

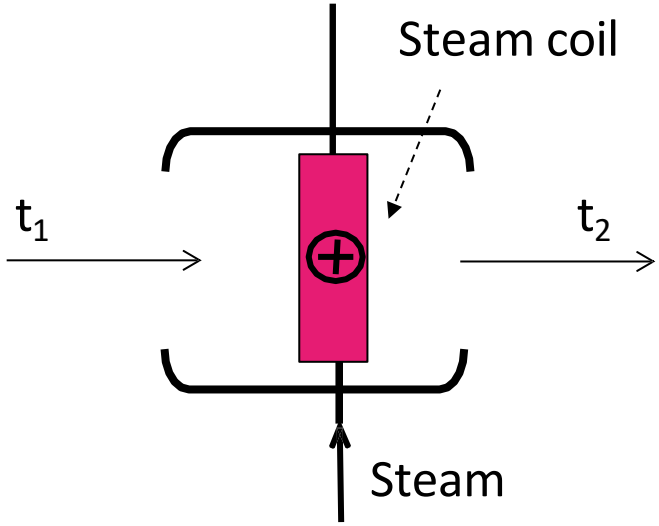
Heating



Heating

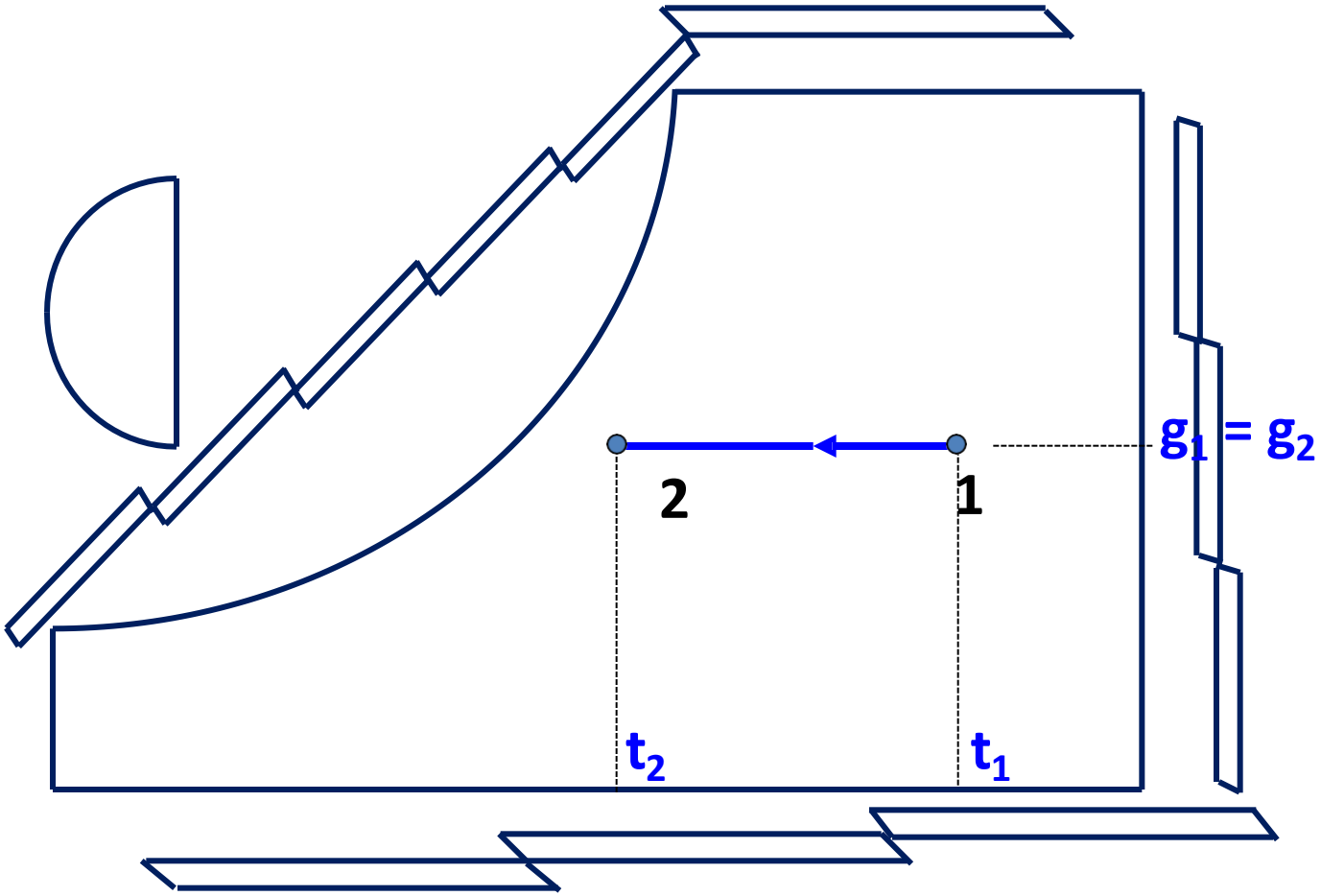

Electric
heater
t_1
t_2
Current


Hot water coil
t_1
t_2
Hot
water

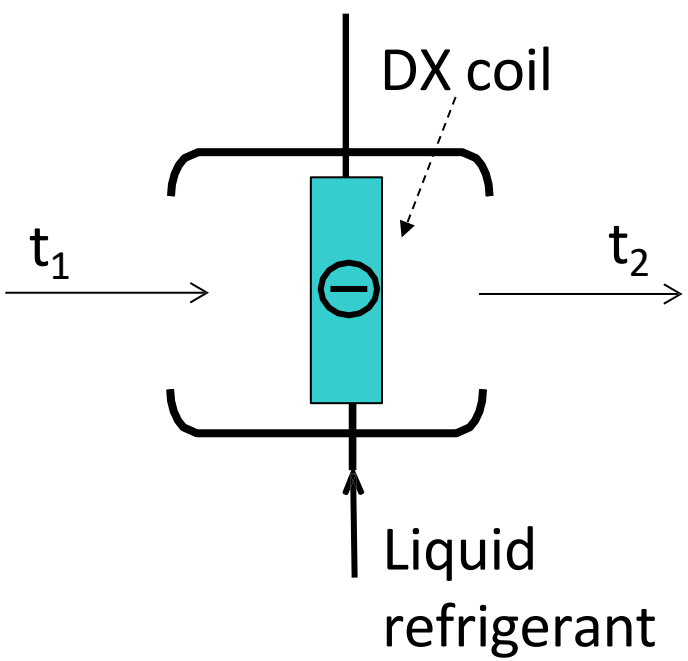

Steam coil
t_1
t_2
Steam

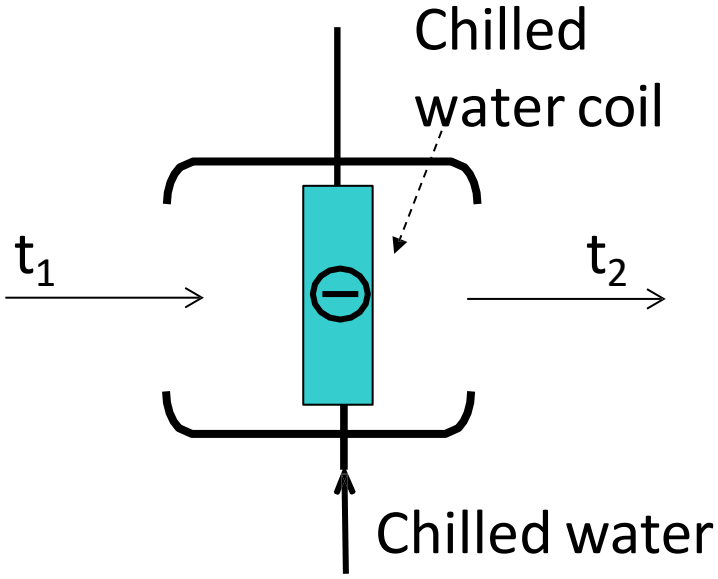


Cooling

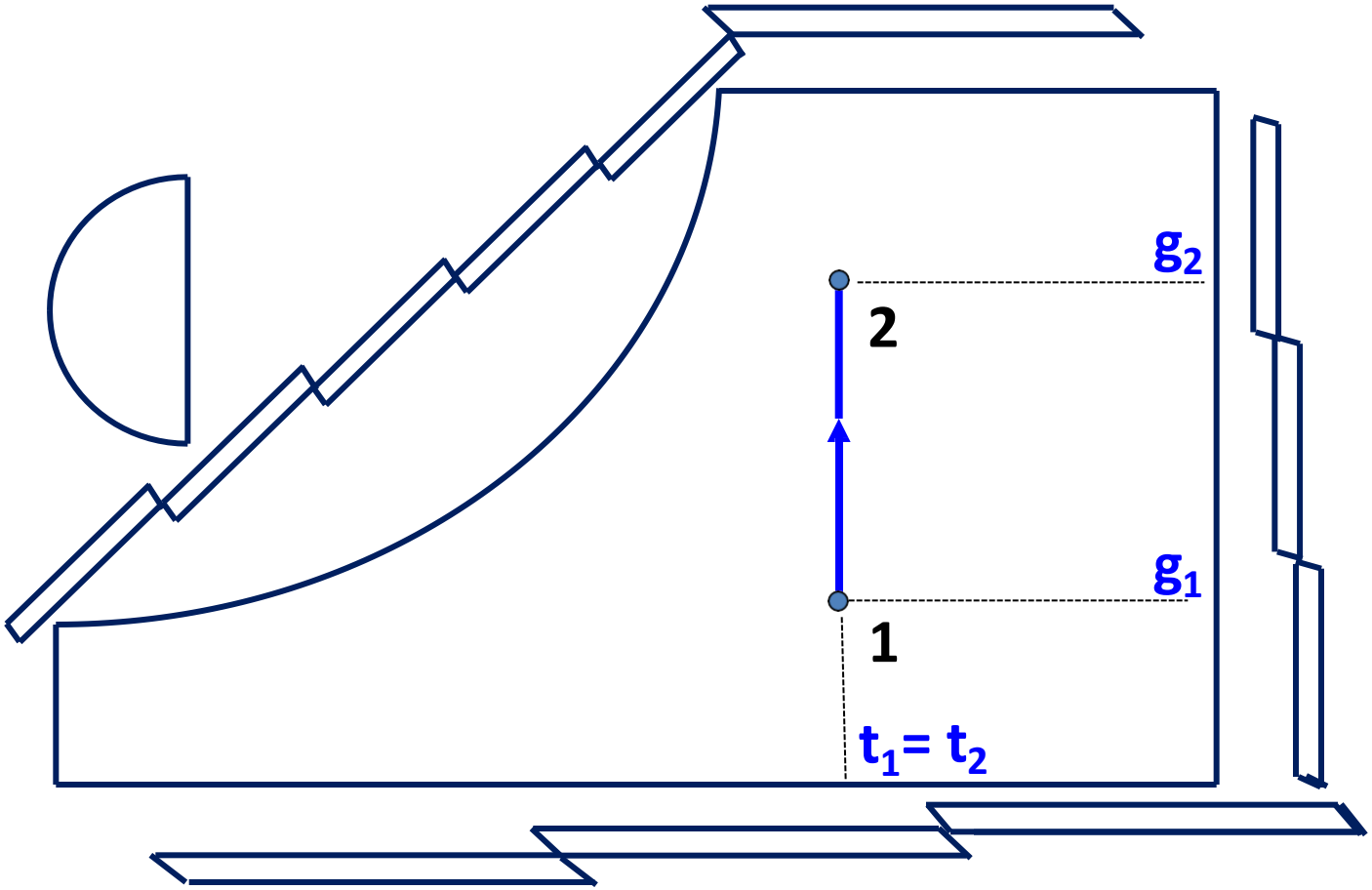


Cooling

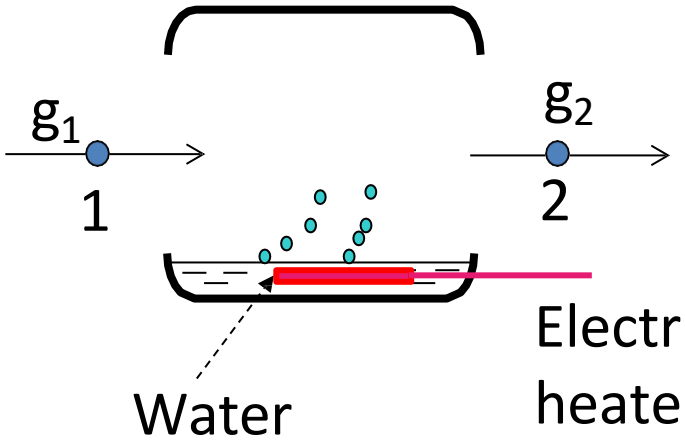

DX coil
t_1
t_2
Liquid
refrigerant

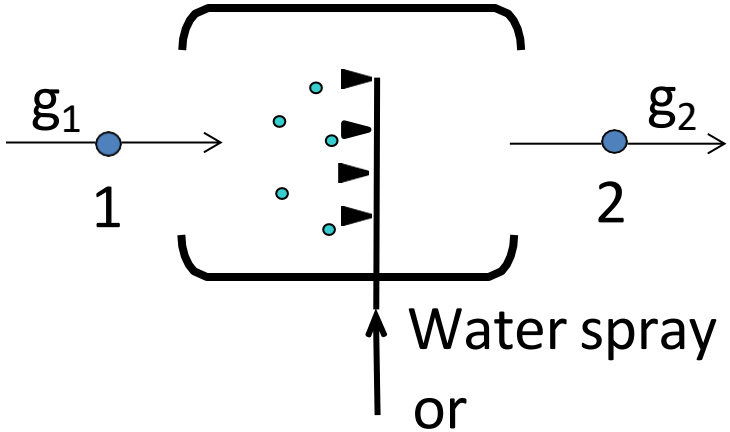

Chilled
water coil
t_1
t_2
Chilled water

Humidifying



Humidification





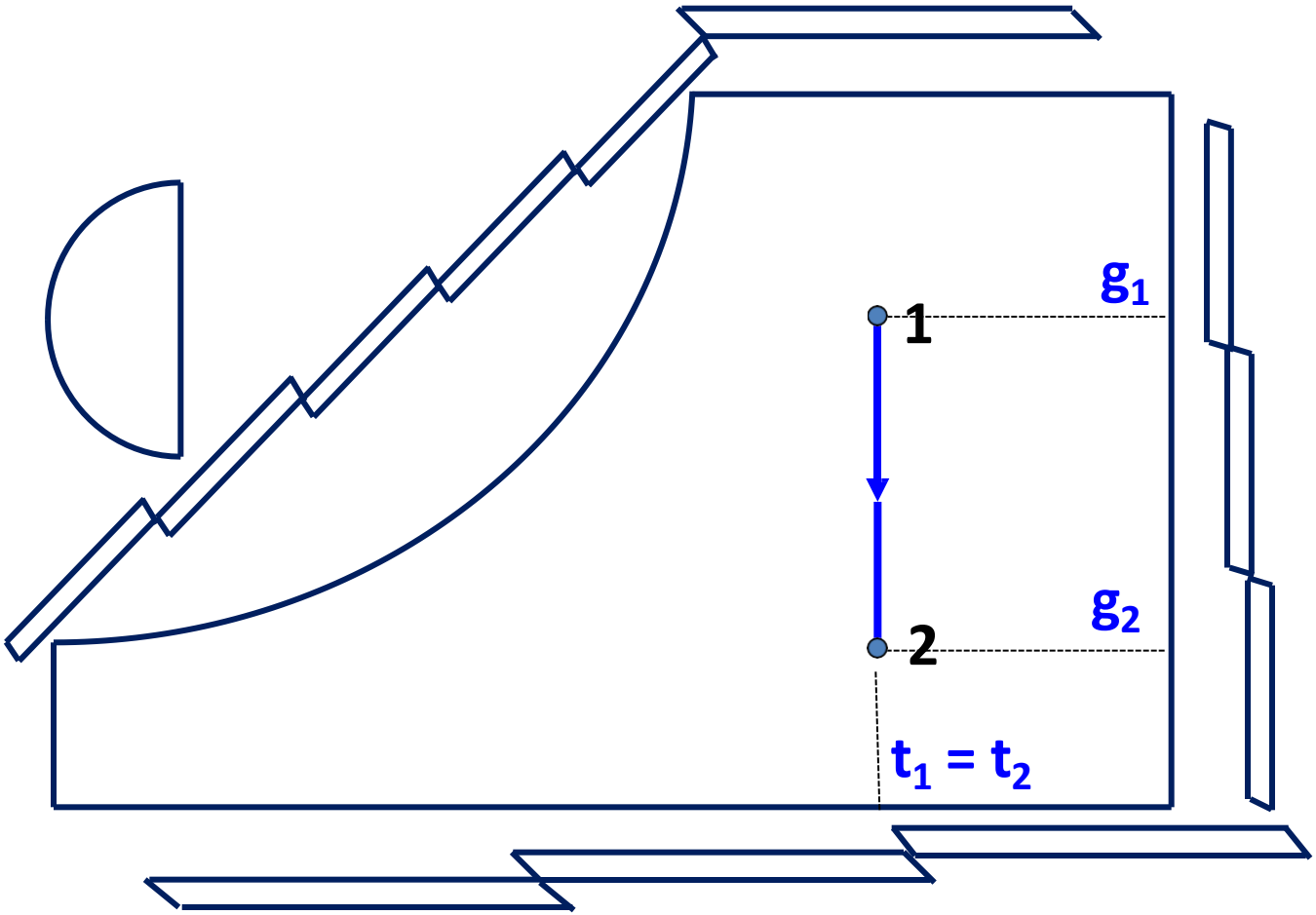
or

Ultrasonic mister

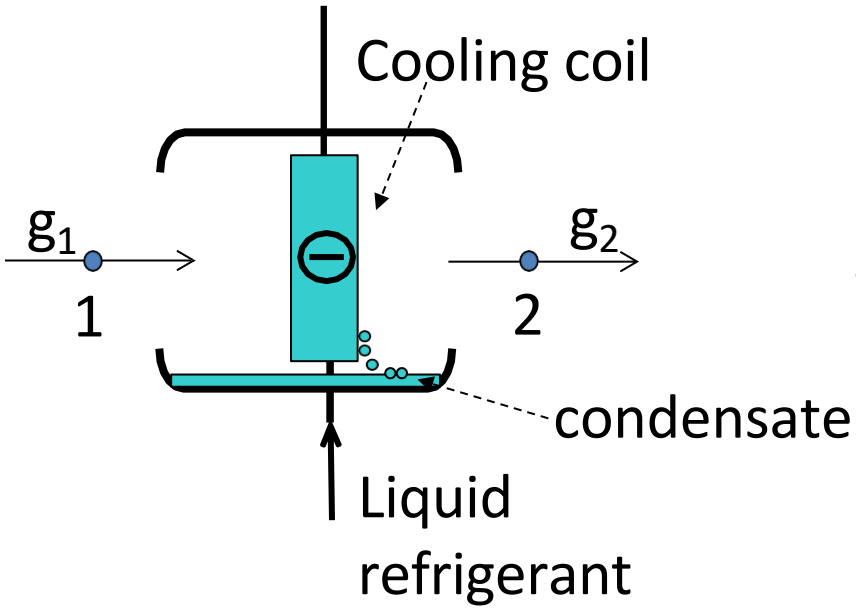
or

Steam injection

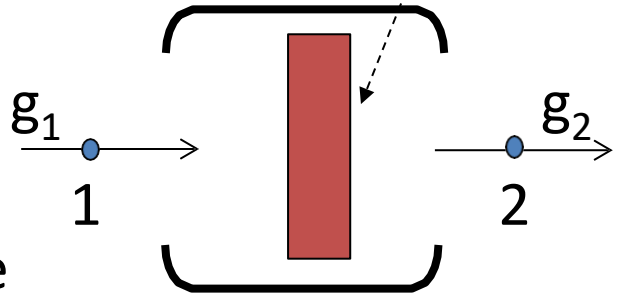
Dehumidifying



Dehumidification



Dessicant wheel (e.g. lithium chloride



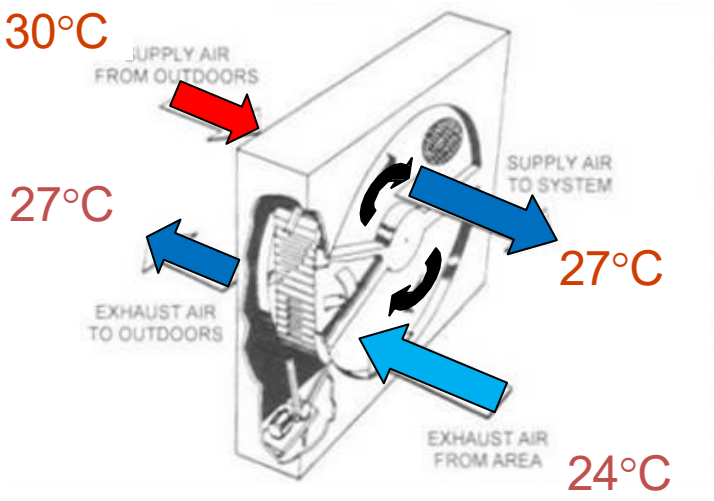




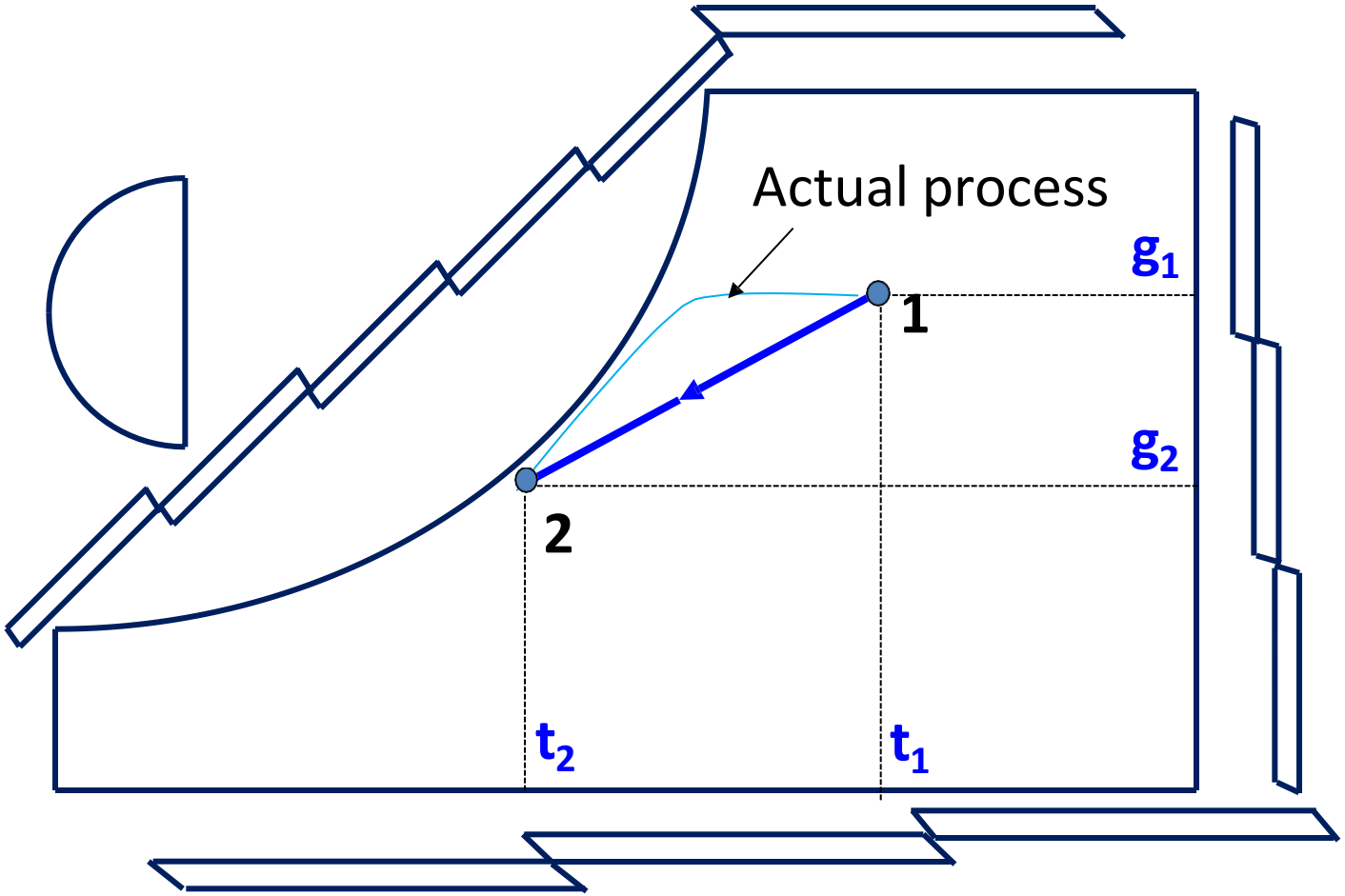
or

Chilled water

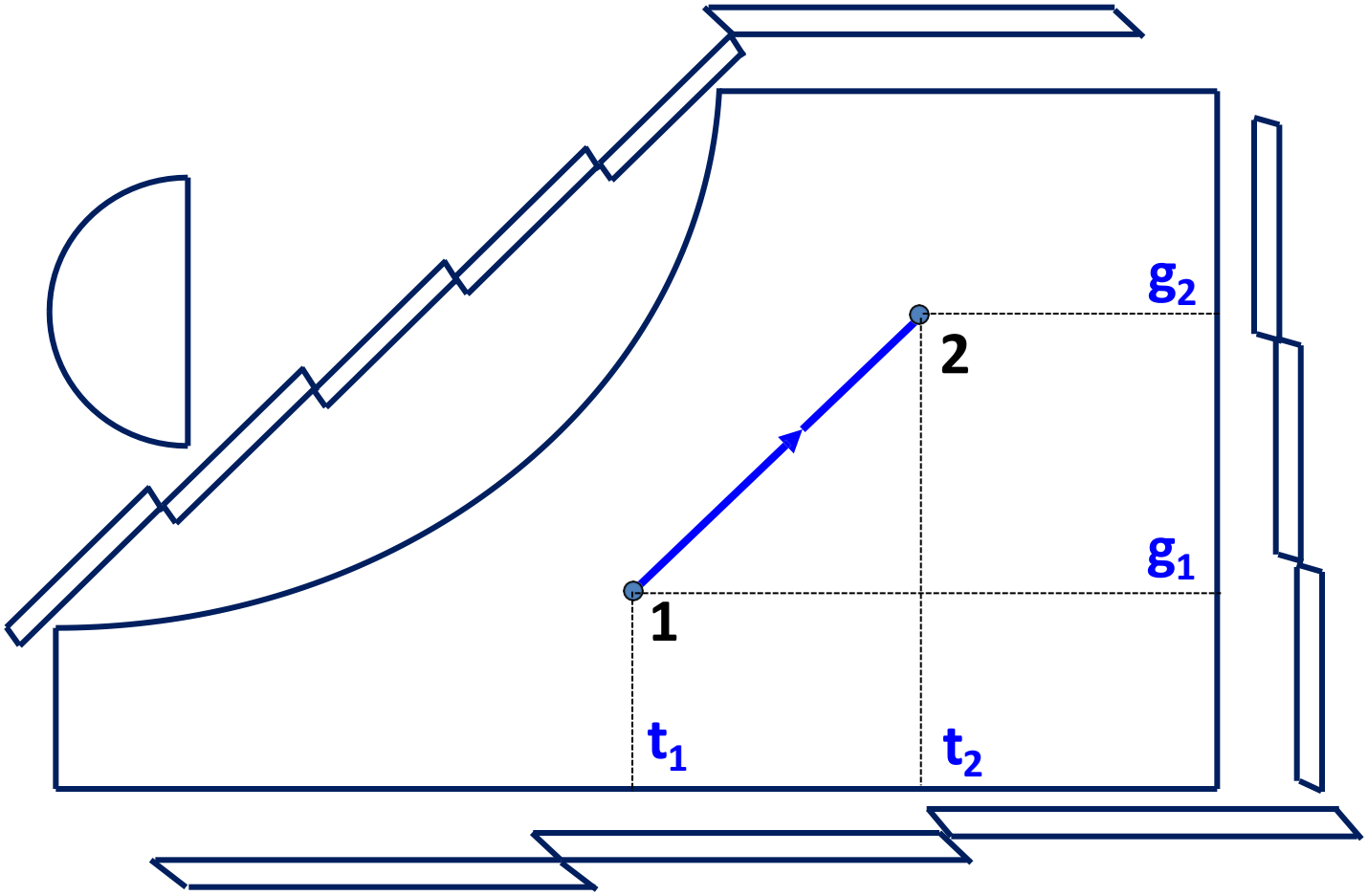




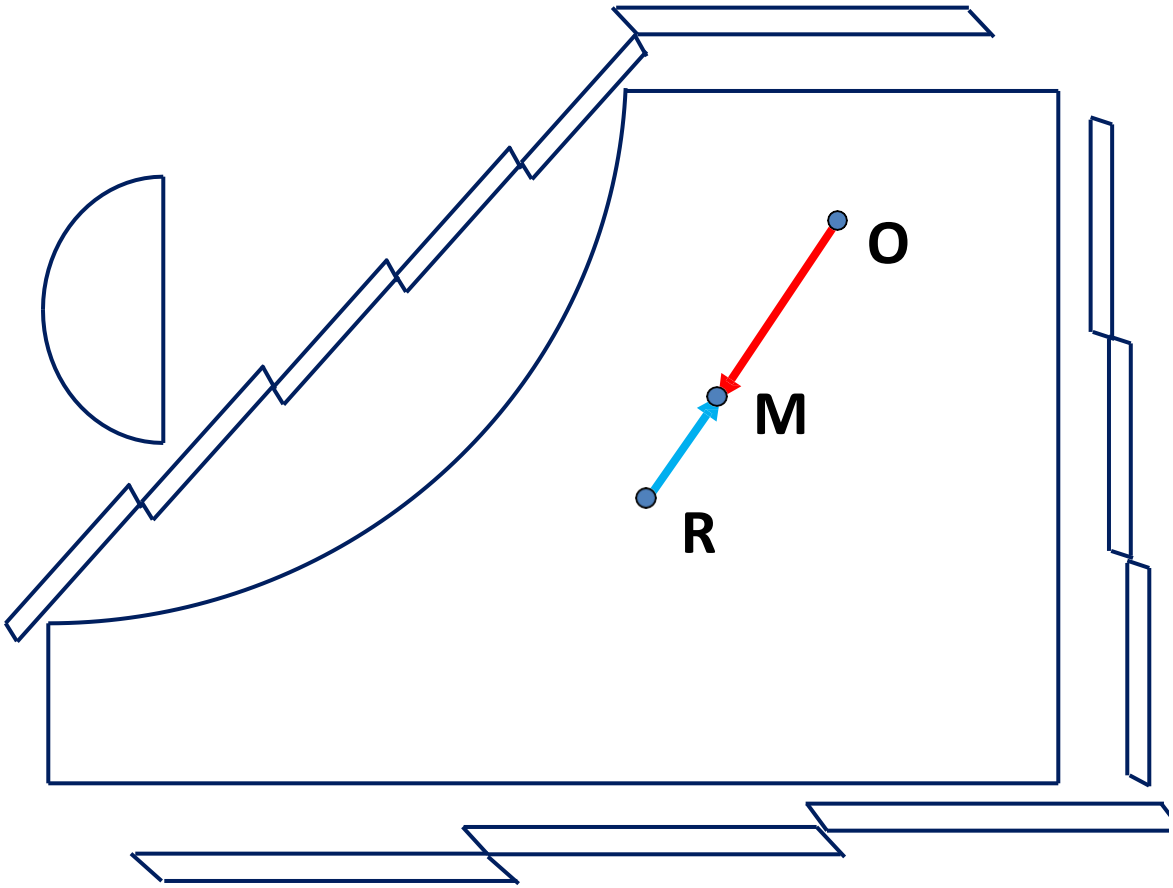
Cooling & Dehumidification

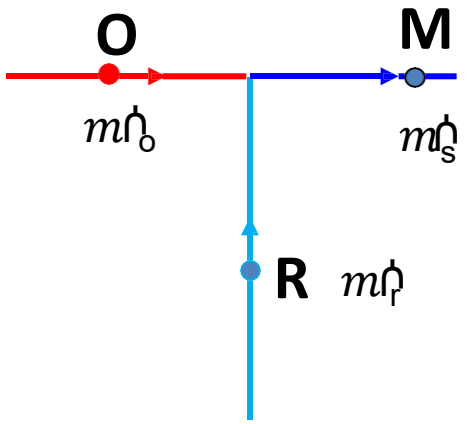


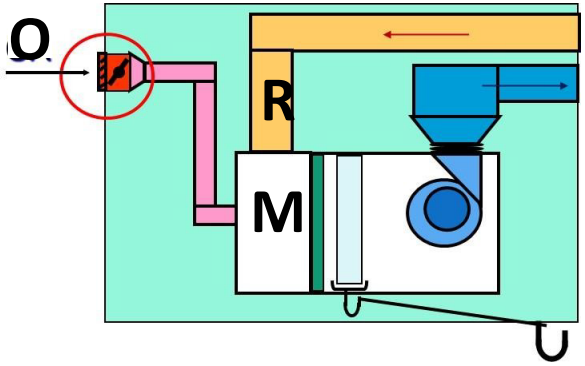
Heating & Humidification



Mixing (Ventilating)







- M lies on a straight line joining O and R
- Where on the line depends on $\dot{m}_o$ and $\dot{m}_r$ and hence, $\dot{m}_s$